Notation and Terminology

$\mathbb{N} =$ the set of all positive integers.

$\mathbb{Z} =$ the set of all integers.

$\mathbb{Q} =$ the set of all rational numbers.

$\mathbb{R} =$ the set of all real numbers.

$\mathbb{R}^n =$ the n-dimensional Euclidean space.

$\mathbb{C} =$ the set of all complex numbers.

$M_n(\mathbb{R})$ = the real vector space of all $n \times n$ matrices with entries in $\mathbb{R}$.

$M_n(\mathbb{C})$ = the complex vector space of all $n \times n$ matrices with entries in $\mathbb{C}$.

$\gcd(m, n)$ = the greatest common divisor of the integers m and n.

$M^{\top} =$ the transpose of the matrix M.

$A - B =$ the complement of the set B in the set A, that is, $\{x \in A : x \notin B\}$.

$\ln x =$ the natural logarithm of x (to the base e).

$|x| =$ the absolute value of x.

$y', y'', y''' =$ the first, second and the third derivatives of the function y, respectively.

$S_n =$ the symmetric group consisting of all permutations of $\{1,2, \dots, n\}$.

$\mathbb{Z}_n =$ the additive group of integers modulo n.

$f \circ g$ is the composite function defined by $(f \circ g)(x) = f(g(x))$.

The phrase '**real vector space**' refers to a vector space over $\mathbb{R}$.

Section A: Q.1 – Q.10 carry ONE mark each.	
Q.1	Consider the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$. If the eighth power of M satisfies $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then the value of x is
(A)	21
(B)	22
(C)	34
(D)	35
Q.2	The rank of the 4×6 matrix $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ with entries in $\mathbb{R}$, is
(A)	1
(B)	2
(C)	3
(D)	4

Q.3	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 6, together with the zero polynomial. Then which one of the following is true?
(A)	$\{f \in V : f(1/2) \notin \mathbb{Q}\}$ is a subspace of V.
(B)	$\{f \in V : f(1/2) = 1\}$ is a subspace of V.
(C)	$\{f \in V : f(1/2) = f(1)\}$ is a subspace of V.
(D)	$\{f \in V : f'(1/2) = 1\}$ is a subspace of V.
Q.4	Let G be a group of order 2022. Let H and K be subgroups of G of order 337 and 674, respectively. If $H \cup K$ is also a subgroup of G, then which one of the following is FALSE?
(A)	H is a normal subgroup of $H \cup K$.
(B)	The order of $H \cup K$ is 1011.
(C)	The order of $H \cup K$ is 674.
(D)	K is a normal subgroup of $H \cup K$.

Q.5	The radius of convergence of the power series $$\sum_{n=1}^{\infty}\left(\frac{n^3}{4^n}\right)x^{5n}$$ is
(A)	4
(B)	$\sqrt[5]{4}$
(C)	$\frac{1}{4}$
(D)	$\frac{1}{\sqrt[5]{4}}$

Q.6	Let (x_n) and (y_n) be sequences of real numbers defined by $$x_1 = 1,\quad y_1 = \frac{1}{2},\quad x_{n+1} = \frac{x_n + y_n}{2},\quad \text{and} \quad y_{n+1} = \sqrt{x_n y_n} \quad \text{for all } n \in \mathbb{N}.$$ Then which one of the following is true?
(A)	(x_n) is convergent, but (y_n) is not convergent.
(B)	(x_n) is not convergent, but (y_n) is convergent.
(C)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n > \lim_{n\to\infty} y_n$.
(D)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n$.
Q.7	Suppose $$a_n = \frac{3^n + 3}{5^n - 5} \quad \text{and} \quad b_n = \frac{1}{(1+n^2)^{\frac{1}{4}}} \quad \text{for } n = 2,3,4,\dots.$$ Then which one of the following is true?
(A)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are convergent.
(B)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are divergent.
(C)	$\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is divergent.
(D)	$\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is convergent.

Q.8	Consider the series $$\sum_{n=1}^{\infty} \frac{1}{n^m\left(1+\frac{1}{n^p}\right)}$$ where m and p are real numbers. Under which of the following conditions does the above series converge?
(A)	$m > 1$.
(B)	$0 < m < 1$ and $p > 1$.
(C)	$0 < m \leq 1$ and $0 \leq p \leq 1$.
(D)	$m = 1$ and $p > 1$.

Q.9	Let c be a positive real number and let $u: \mathbb{R}^2 \to \mathbb{R}$ be defined by $$u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} e^{s^2} ds \quad \text{for} \quad (x,t) \in \mathbb{R}^2.$$ Then which one of the following is true?
(A)	$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ on $\mathbb{R}^2$.
(B)	$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ on $\mathbb{R}^2$.
(C)	$\frac{\partial u}{\partial t}\frac{\partial u}{\partial x} = 0$ on $\mathbb{R}^2$.
(D)	$\frac{\partial^2 u}{\partial t\, \partial x} = 0$ on $\mathbb{R}^2$.

Q.10	Let $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Consider the functions $$u : \mathbb{R}^2 - \{(0,0)\} \to \mathbb{R} \quad \text{and} \quad v : \mathbb{R}^2 - \{(0,0)\} \to \mathbb{R}$$ given by $$u(x,y) = x - \frac{x}{x^2+y^2} \quad \text{and} \quad v(x,y) = y + \frac{y}{x^2+y^2}.$$ The value of the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ at the point $(\cos\theta, \sin\theta)$ is equal to
(A)	$4\sin\theta$.
(B)	$4\cos\theta$.
(C)	$4\sin^2\theta$.
(D)	$4\cos^2\theta$.

Section A: Q.11 – Q.30 Carry TWO marks each.	
Q.11	Consider the open rectangle $G = \{(s,t) \in \mathbb{R}^2 : \ 0 < s < 1 \text{ and } 0 < t < 1\}$ and the map $T: G \to \mathbb{R}^2$ given by $$T(s,t) = \left(\frac{\pi s(1-t)}{2}, \ \frac{\pi(1-s)}{2}\right) \quad \text{for } (s,t) \in G .$$ Then the area of the image $T(G)$ of the map T is equal to
(A)	$\frac{\pi}{4}$
(B)	$\frac{\pi^2}{4}$
(C)	$\frac{\pi^2}{8}$
(D)	1

Q.12	Let T denote the sum of the convergent series $$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\cdots+\frac{(-1)^{n+1}}{n}+\cdots$$ and let S denote the sum of the convergent series $$1-\frac{1}{2}-\frac{1}{4}+\frac{1}{3}-\frac{1}{6}-\frac{1}{8}+\frac{1}{5}-\frac{1}{10}-\frac{1}{12}+\cdots=\sum_{n=1}^{\infty}a_n,$$ where $$a_{3m-2}=\frac{1}{2m-1},\quad a_{3m-1}=\frac{-1}{4m-2}\quad \text{and}\quad a_{3m}=\frac{-1}{4m}\quad \text{for } m\in\mathbb{N}.$$ Then which one of the following is true?
(A)	$T=S$ and $S\neq 0$.
(B)	$2T=S$ and $S\neq 0$.
(C)	$T=2S$ and $S\neq 0$.
(D)	$T=S=0$.

Q.13	Let $u: \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that $u(0) > 0$ and $u'(0) > 0$. Suppose u satisfies $$u''(x) = \frac{u(x)}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$ Consider the following two statements: I. The function uu' is monotonically increasing on $[0, \infty)$. II. The function u is monotonically increasing on $[0, \infty)$. Then which one of the following is correct?
(A)	Both I and II are false.
(B)	Both I and II are true.
(C)	I is false, but II is true.
(D)	I is true, but II is false.

Q.14	The value of $$\lim_{n\to\infty}\sum_{k=2}^{n}\frac{\sqrt{n+1}-\sqrt{n}}{k(\ln k)^2}$$ is equal to
(A)	∞
(B)	1
(C)	e
(D)	0

Q.15	For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t. Define functions $h: \mathbb{R}^2 \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ by $$h(x,y) = \begin{cases} \frac{-1}{x^2-y} & \text{if } x^2 \neq y, \\ 0 & \text{if } x^2 = y \end{cases} \quad \text{and} \quad g(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$ Then which one of the following is FALSE?
(A)	$\lim_{(x,y)\to(\sqrt{2},\pi)} \cos\left(\frac{x^2 y}{x^2+1}\right) = \frac{-1}{2}$.
(B)	$\lim_{(x,y)\to(\sqrt{2},2)} e^{h(x,y)} = 0$.
(C)	$\lim_{(x,y)\to(e,e)} \ln\left(x^{y-[y]}\right) = e - 2$.
(D)	$\lim_{(x,y)\to(0,0)} e^{2y} g(x) = 1$.

Q.16	Let $P \in M_4(\mathbb{R})$ be such that P^4 is the zero matrix, but P^3 is a nonzero matrix. Then which one of the following is FALSE?
(A)	For every nonzero vector $v \in \mathbb{R}^4$, the subset $\{v, Pv, P^2v, P^3v\}$ of the real vector space $\mathbb{R}^4$ is linearly independent.
(B)	The rank of P^k is $4-k$ for every $k \in \{1,2,3,4\}$.
(C)	0 is an eigenvalue of P.
(D)	If $Q \in M_4(\mathbb{R})$ is such that Q^4 is the zero matrix, but Q^3 is a nonzero matrix, then there exists a nonsingular matrix $S \in M_4(\mathbb{R})$ such that $S^{-1}QS = P$.

Q.17	For $X, Y \in M_2(\mathbb{R})$, define $(X, Y) = XY - YX$. Let $\mathbf{0} \in M_2(\mathbb{R})$ denote the zero matrix. Consider the two statements: $P:\ \ \big(X,(Y,Z)\big) + \big(Y,(Z,X)\big) + \big(Z,(X,Y)\big) = \mathbf{0}$ for all $X, Y, Z \in M_2(\mathbb{R})$. $Q:\ \ \big(X,(Y,Z)\big) = \big((X,Y),Z\big)$ for all $X, Y, Z \in M_2(\mathbb{R})$. Then which one of the following is correct?
(A)	Both P and Q are true.
(B)	P is true, but Q is false.
(C)	P is false, but Q is true.
(D)	Both P and Q are false.

Q.18	Consider the system of linear equations $$\begin{aligned} x+y+t &= 4,\\ 2x-4t &= 7,\\ x+y+z &= 5,\\ x-3y-z-10t &= \lambda, \end{aligned}$$ where x, y, z, t are variables and λ is a constant. Then which one of the following is true?
(A)	If $\lambda = 1$, then the system has a unique solution.
(B)	If $\lambda = 2$, then the system has infinitely many solutions.
(C)	If $\lambda = 1$, then the system has infinitely many solutions.
(D)	If $\lambda = 2$, then the system has a unique solution.
Q.19	Consider the group $(\mathbb{Q}, +)$ and its subgroup $(\mathbb{Z}, +)$. For the quotient group $\mathbb{Q}/\mathbb{Z}$, which one of the following is FALSE?
(A)	$\mathbb{Q}/\mathbb{Z}$ contains a subgroup isomorphic to $(\mathbb{Z}, +)$.
(B)	There is exactly one group homomorphism from $\mathbb{Q}/\mathbb{Z}$ to $(\mathbb{Q}, +)$.
(C)	For all $n \in \mathbb{N}$, there exists $g \in \mathbb{Q}/\mathbb{Z}$ such that the order of g is n.
(D)	$\mathbb{Q}/\mathbb{Z}$ is not a cyclic group.

Q.20	For $P \in M_5(\mathbb{R})$ and $i,j \in \{1,2,\ldots,5\}$, let p_{ij} denote the $(i,j)^{\text{th}}$ entry of P. Let $$S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs} \text{ for } i,j,r,s \in \{1,2,\ldots,5\} \text{ with } i+r = j+s\}.$$ Then which one of the following is FALSE?
(A)	S is a subspace of the vector space over $\mathbb{R}$ of all 5×5 symmetric matrices.
(B)	The dimension of S over $\mathbb{R}$ is 5.
(C)	The dimension of S over $\mathbb{R}$ is 11.
(D)	If $P \in S$ and all the entries of P are integers, then 5 divides the sum of all the diagonal entries of P.

Q.21	On the open interval $(-c, c)$, where c is a positive real number, $y(x)$ is an infinitely differentiable solution of the differential equation $$\frac{dy}{dx} = y^2 - 1 + \cos x,$$ with the initial condition $y(0) = 0$. Then which one of the following is correct?
(A)	$y(x)$ has a local maximum at the origin.
(B)	$y(x)$ has a local minimum at the origin.
(C)	$y(x)$ is strictly increasing on the open interval $(-\delta, \delta)$ for some positive real number δ.
(D)	$y(x)$ is strictly decreasing on the open interval $(-\delta, \delta)$ for some positive real number δ.

Q.22	Let $H : \mathbb{R} \to \mathbb{R}$ be the function given by $H(x) = \frac{1}{2}(e^x + e^{-x})$ for $x \in \mathbb{R}$. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $$f(x) = \int_0^{\pi} H(x \sin\theta) d\theta \quad \text{for } x \in \mathbb{R}.$$ Then which one of the following is true?
(A)	$xf''(x) + f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(B)	$xf''(x) - f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(C)	$xf''(x) + f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.
(D)	$xf''(x) - f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.

Q.23	Consider the differential equation $$y'' + ay' + y = \sin x \quad \text{for } x \in \mathbb{R}. \qquad (**)$$ Then which one of the following is true?
(A)	If $a = 0$, then all the solutions of $(**)$ are unbounded over $\mathbb{R}$.
(B)	If $a = 1$, then all the solutions of $(**)$ are unbounded over $(0, \infty)$.
(C)	If $a = 1$, then all the solutions of $(**)$ tend to zero as $x \to \infty$.
(D)	If $a = 2$, then all the solutions of $(**)$ are bounded over $(-\infty, 0)$.
Q.24	For $g \in \mathbb{Z}$, let $\overline{g} \in \mathbb{Z}_{37}$ denote the residue class of g modulo 37. Consider the group $U_{37} = \{\overline{g} \in \mathbb{Z}_{37} : 1 \leq g \leq 37 \text{ with } \gcd(g, 37) = 1\}$ with respect to multiplication modulo 37. Then which one of the following is FALSE?
(A)	The set $\{\overline{g} \in U_{37} : \overline{g} = (\overline{g})^{-1}\}$ contains exactly 2 elements.
(B)	The order of the element $\overline{10}$ in U_{37} is 36.
(C)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Z}, +)$.
(D)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Q}, +)$.

Q.25	For some real number c with $0 < c < 1$, let $\phi: (1-c, 1+c) \to (0, \infty)$ be a differentiable function such that $\phi(1) = 1$ and $y = \phi(x)$ is a solution of the differential equation $$(x^2 + y^2)dx - 4xy\, dy = 0.$$ Then which one of the following is true?
(A)	$\left(3(\phi(x))^2 + x^2\right)^2 = 4x.$
(B)	$\left(3(\phi(x))^2 - x^2\right)^2 = 4x.$
(C)	$\left(3(\phi(x))^2 + x^2\right)^2 = 4\phi(x).$
(D)	$\left(3(\phi(x))^2 - x^2\right)^2 = 4\phi(x).$

Q.26	For a 4×4 matrix $M \in M_4(\mathbb{C})$, let $\bar{M}$ denote the matrix obtained from M by replacing each entry of M by its complex conjugate. Consider the real vector space $$H = \{M \in M_4(\mathbb{C}) : \quad M^\top = \bar{M}\}$$ where $M^\top$ denotes the transpose of M. The dimension of H as a vector space over $\mathbb{R}$ is equal to
(A)	6
(B)	16
(C)	15
(D)	12

Q.27	Let a, b be positive real numbers such that $a < b$. Given that $$\lim_{N\to\infty}\int_0^N e^{-t^2}dt = \frac{\sqrt{\pi}}{2},$$ the value of $$\lim_{N\to\infty}\int_0^N \frac{1}{t^2}\left(e^{-at^2} - e^{-bt^2}\right)dt$$ is equal to
(A)	$\sqrt{\pi}(\sqrt{a} - \sqrt{b})$.
(B)	$\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(C)	$-\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(D)	$\sqrt{\pi}(\sqrt{b} - \sqrt{a})$.

Q.28	For $-1 \le x \le 1$, if $f(x)$ is the sum of the convergent power series $$x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \cdots + \frac{x^n}{n^2} + \cdots$$ then $f\left(\frac{1}{2}\right)$ is equal to
(A)	$\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\,dt.$
(B)	$-\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\,dt.$
(C)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1+t)\,dt.$
(D)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1-t)\,dt.$

Q.29	For $n \in \mathbb{N}$ and $x \in [1, \infty)$, let $$f_n(x) = \int_0^{\pi} \left(x^2 + (\cos\theta)\sqrt{x^2 - 1}\right)^n d\theta.$$ Then which one of the following is true?
(A)	$f_n(x)$ is not a polynomial in x if n is odd and $n \geq 3$.
(B)	$f_n(x)$ is not a polynomial in x if n is even and $n \geq 4$.
(C)	$f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
(D)	$f_n(x)$ is not a polynomial in x for any $n \geq 3$.
Q.30	Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3, respectively. Then the entry in the first row and the third column of P is
(A)	0
(B)	1
(C)	-1
(D)	2

Section B: Q.31 – Q.40 Carry TWO marks each.	
Q.31	Let $(-c, c)$ be the largest open interval in $\mathbb{R}$ (where c is either a positive real number or $c = \infty$) on which the solution $y(x)$ of the differential equation $$\frac{dy}{dx} = x^2 + y^2 + 1 \quad \text{with initial condition} \quad y(0) = 0$$ exists and is unique. Then which of the following is/are true?
(A)	$y(x)$ is an odd function on $(-c, c)$.
(B)	$y(x)$ is an even function on $(-c, c)$.
(C)	$\big(y(x)\big)^2$ has a local minimum at 0.
(D)	$\big(y(x)\big)^2$ has a local maximum at 0.

Q.32	Let S be the set of all continuous functions $f: [-1,1] \to \mathbb{R}$ satisfying the following three conditions: (i) f is infinitely differentiable on the open interval $(-1,1)$, (ii) the Taylor series $$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots$$ of f at 0 converges to $f(x)$ for each $x \in (-1,1)$, (iii) $f\left(\frac{1}{n}\right) = 0$ for all $n \in \mathbb{N}$. Then which of the following is/are true?
(A)	$f(0) = 0$ for every $f \in S$.
(B)	$f'\left(\frac{1}{2}\right) = 0$ for every $f \in S$.
(C)	There exists $f \in S$ such that $f'\left(\frac{1}{2}\right) \neq 0$.
(D)	There exists $f \in S$ such that $f(x) \neq 0$ for some $x \in [-1,1]$.

Q.33	Define $f:[0,1] \to [0,1]$ by $$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \text{ for some } m,n \in \mathbb{N} \text{ with } m \leq n \text{ and } \gcd(m,n) = 1, \\ 0 & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$ and define $g:[0,1] \to [0,1]$ by $$g(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \in (0,1]. \end{cases}$$ Then which of the following is/are true?
(A)	f is Riemann integrable on $[0,1]$.
(B)	g is Riemann integrable on $[0,1]$.
(C)	The composite function $f \circ g$ is Riemann integrable on $[0,1]$.
(D)	The composite function $g \circ f$ is Riemann integrable on $[0,1]$.

Q.34	Let S be the set of all functions $f: \mathbb{R} \to \mathbb{R}$ satisfying $$\|f(x) - f(y)\|^2 \leq \|x - y\|^3 \quad \text{for all } x, y \in \mathbb{R}.$$ Then which of the following is/are true?
(A)	Every function in S is differentiable.
(B)	There exists a function $f \in S$ such that f is differentiable, but f is not twice differentiable.
(C)	There exists a function $f \in S$ such that f is twice differentiable, but f is not thrice differentiable.
(D)	Every function in S is infinitely differentiable.

Q.35	A real-valued function $y(x)$ defined on $\mathbb{R}$ is said to be **periodic** if there exists a real number $T > 0$ such that $y(x+T) = y(x)$ for all $x \in \mathbb{R}$. Consider the differential equation $$\frac{d^2y}{dx^2} + 4y = \sin(ax), \quad x \in \mathbb{R}, \qquad (*)$$ where $a \in \mathbb{R}$ is a constant. Then which of the following is/are true?
(A)	All solutions of $(*)$ are periodic for every choice of a.
(B)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{R} - \{-2, 2\}$.
(C)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{Q} - \{-2, 2\}$.
(D)	If $a \in \mathbb{R} - \mathbb{Q}$, then there is a unique periodic solution of $(*)$.

Q.36	Let M be a positive real number and let $u, v : \mathbb{R}^2 \to \mathbb{R}$ be continuous functions satisfying $$\sqrt{u(x,y)^2 + v(x,y)^2} \geq M\sqrt{x^2 + y^2} \quad \text{for all } (x,y) \in \mathbb{R}^2.$$ Let $F: \mathbb{R}^2 \to \mathbb{R}^2$ be given by $$F(x,y) = \big(u(x,y), v(x,y)\big) \quad \text{for } (x,y) \in \mathbb{R}^2.$$ Then which of the following is/are true?
(A)	F is injective.
(B)	If K is open in $\mathbb{R}^2$, then $F(K)$ is open in $\mathbb{R}^2$.
(C)	If K is closed in $\mathbb{R}^2$, then $F(K)$ is closed in $\mathbb{R}^2$.
(D)	If E is closed and bounded in $\mathbb{R}^2$, then $F^{-1}(E)$ is closed and bounded in $\mathbb{R}^2$.

Q.37	Let G be a finite group of order at least two and let e denote the identity element of G. Let $\sigma: G \to G$ be a bijective group homomorphism that satisfies the following two conditions: (i) If $\sigma(g) = g$ for some $g \in G$, then $g = e$, (ii) $(\sigma \circ \sigma)(g) = g$ for all $g \in G$. Then which of the following is/are correct?
(A)	For each $g \in G$, there exists $h \in G$ such that $h^{-1}\sigma(h) = g$.
(B)	There exists $x \in G$ such that $x\sigma(x) \neq e$.
(C)	The map σ satisfies $\sigma(x) = x^{-1}$ for every $x \in G$.
(D)	The order of the group G is an odd number.

Q.38	Let (x_n) be a sequence of real numbers. Consider the set $$P = \{n \in \mathbb{N} :\ x_n >\ x_m \text{ for all } m \in \mathbb{N} \text{ with } m > n\}.$$ Then which of the following is/are true?
(A)	If P is finite, then (x_n) has a monotonically increasing subsequence.
(B)	If P is finite, then no subsequence of (x_n) is monotonically increasing.
(C)	If P is infinite, then (x_n) has a monotonically decreasing subsequence.
(D)	If P is infinite, then no subsequence of (x_n) is monotonically decreasing.
Q.39	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 5, together with the zero polynomial. Let $T: V \to \mathbb{R}$ be the linear map defined by $T(1) = 1$ and $$T\big(x(x-1)\cdots(x-k+1)\big) = 1 \quad \text{for } 1 \le k \le 5.$$ Then which of the following is/are true?
(A)	$T(x^4) = 15$.
(B)	$T(x^3) = 5$.
(C)	$T(x^4) = 14$.
(D)	$T(x^3) = 3$.

Q.40	Let P be a fixed 3×3 matrix with entries in $\mathbb{R}$. Which of the following maps from $M_3(\mathbb{R})$ to $M_3(\mathbb{R})$ is/are linear?
(A)	$T_1: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_1(M) = MP - PM$ for $M \in M_3(\mathbb{R})$.
(B)	$T_2: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_2(M) = M^2P - P^2M$ for $M \in M_3(\mathbb{R})$.
(C)	$T_3: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_3(M) = MP^2 + P^2M$ for $M \in M_3(\mathbb{R})$.
(D)	$T_4: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_4(M) = MP^2 - PM^2$ for $M \in M_3(\mathbb{R})$.

Section C: Q.41 – Q.50 Carry ONE mark each.

Q.41	The value of the limit $$\lim_{n\to\infty}\left(\frac{(1^4+2^4+\cdots+n^4)}{n^5}+\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{n+1}}+\frac{1}{\sqrt{n+2}}+\cdots+\frac{1}{\sqrt{4n}}\right)\right)$$ is equal to __________. (*Rounded off to two decimal places*)

Q.42	Consider the function $u: \mathbb{R}^3 \to \mathbb{R}$ given by $$u(x_1, x_2, x_3) = x_1 x_2^4 x_3^2 - x_1^3 x_3^4 - 26 x_1^2 x_2^2 x_3^3.$$ Let $c \in \mathbb{R}$ and $k \in \mathbb{N}$ be such that $$x_1 \frac{\partial u}{\partial x_2} + 2x_2 \frac{\partial u}{\partial x_3}$$ evaluated at the point (t, t^2, t^3), equals ct^k for every $t \in \mathbb{R}$. Then the value of k is equal to ___________.
Q.43	Let $y(x)$ be the solution of the differential equation $$\frac{dy}{dx} + 3x^2 y = x^2, \qquad \text{for } x \in \mathbb{R},$$ satisfying the initial condition $y(0) = 4$. Then $\lim_{x \to \infty} y(x)$ is equal to ___________. (*Rounded off to two decimal places*)
Q.44	The sum of the series $$\sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)}$$ is equal to ___________. (*Rounded off to two decimal places*)
Q.45	The number of distinct subgroups of $\mathbb{Z}_{999}$ is ___________.

Q.46	The number of elements of order 12 in the symmetric group S_7 is equal to __________.
Q.47	Let $y(x)$ be the solution of the differential equation $$xy^2y' + y^3 = \frac{\sin x}{x} \quad \text{for } x > 0,$$ satisfying $y\left(\frac{\pi}{2}\right) = 0$. Then the value of $y\left(\frac{5\pi}{2}\right)$ is equal to __________. (*Rounded off to two decimal places*)
Q.48	Consider the region $$G = \{(x, y, z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2,\ x^2 + y^2 < 1\}.$$ Then the volume of G is equal to __________. (*Rounded off to two decimal places*)

Q.49	Given that $y(x)$ is a solution of the differential equation $$x^2y'' + xy' - 4y = x^2$$ on the interval $(0, \infty)$ such that $\lim_{x \to 0^+} y(x)$ exists and $y(1) = 1$. The value of $y'(1)$ is equal to __________. (*Rounded off to two decimal places*)
Q.50	Consider the family $\mathcal{F}_1$ of curves lying in the region $$\{(x, y) \in \mathbb{R}^2 : y > 0 \text{ and } 0 < x < \pi\}$$ and given by $$y = \frac{c(1 - \cos x)}{\sin x}, \quad \text{where } c \text{ is a positive real number.}$$ Let $\mathcal{F}_2$ be the family of orthogonal trajectories to $\mathcal{F}_1$. Consider the curve $\mathcal{C}$ belonging to the family $\mathcal{F}_2$ passing through the point $\left(\frac{\pi}{3}, 1\right)$. If a is a real number such that $\left(\frac{\pi}{4}, a\right)$ lies on $\mathcal{C}$, then the value of a^4 is equal to __________. (*Rounded off to two decimal places*)

Section C: Q.51 – Q.60 Carry TWO marks each.

Q.51 For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t.

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$. Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ be defined by $f(0,0) = g(0,0) = 0$ and

$$f(x,y) = [x^2 + y^2]\frac{x^2y^2}{x^4+y^4}, \qquad g(x,y) = [y^2]\frac{xy}{x^2+y^2}$$

for $(x, y) \neq (0,0)$. Let E be the set of points of D at which both f and g are discontinuous. The number of elements in the set E is ________.

Q.52 If G is the region in $\mathbb{R}^2$ given by

$$G = \left\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1, \ \frac{x}{\sqrt{3}} < y < \sqrt{3}x, \ x > 0, \ y > 0\right\}$$

then the value of

$$\frac{200}{\pi}\iint_G x^2 dx\, dy$$

is equal to __________. (*Rounded off to two decimal places*)

Q.53	Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$ and let A^{T} denote the transpose of A. Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be column vectors with entries in $\mathbb{R}$ such that $u_1^2 + u_2^2 = 1$ and $v_1^2 + v_2^2 + v_3^2 = 1$. Suppose $$Au = \sqrt{2}\,v \quad \text{and} \quad A^{\mathsf{T}}v = \sqrt{2}\,u.$$ Then $\lvert u_1 + 2\sqrt{2}\,v_1 \rvert$ is equal to __________. (*Rounded off to two decimal places*)
Q.54	Let $f: [0,\pi] \to \mathbb{R}$ be the function defined by $$f(x) = \begin{cases} (x-\pi)e^{\sin x} & \text{if } 0 \le x \le \dfrac{\pi}{2}, \\ xe^{\sin x} + \dfrac{4}{\pi} & \text{if } \dfrac{\pi}{2} < x \le \pi. \end{cases}$$ Then the value of $$\int_0^{\pi} f(x)\,dx$$ is equal to __________. (*Rounded off to two decimal places*)

Q.55	Let r be the radius of convergence of the power series $$\frac{1}{3}+\frac{x}{5}+\frac{x^2}{3^2}+\frac{x^3}{5^2}+\frac{x^4}{3^3}+\frac{x^5}{5^3}+\frac{x^6}{3^4}+\frac{x^7}{5^4}+\cdots.$$ Then the value of r^2 is equal to ___________. (*Rounded off to two decimal places*)
Q.56	Define $f:\mathbb{R}^2\to\mathbb{R}$ by $$f(x,y)=x^2+2y^2-x \quad \text{for } (x,y)\in\mathbb{R}^2.$$ Let $D=\{(x,y)\in\mathbb{R}^2 : x^2+y^2\le 1\}$ and $E=\left\{(x,y)\in\mathbb{R}^2 : \frac{x^2}{4}+\frac{y^2}{9}\le 1\right\}$. Consider the sets $$D_{\max}=\{(a,b)\in D : f \text{ has absolute maximum on } D \text{ at } (a,b)\},$$ $$D_{\min}=\{(a,b)\in D : f \text{ has absolute minimum on } D \text{ at } (a,b)\},$$ $$E_{\max}=\{(c,d)\in E : f \text{ has absolute maximum on } E \text{ at } (c,d)\},$$ $$E_{\min}=\{(c,d)\in E : f \text{ has absolute minimum on } E \text{ at } (c,d)\}.$$ Then the total number of elements in the set $$D_{\max}\cup D_{\min}\cup E_{\max}\cup E_{\min}$$ is equal to ___________.

Q.57	Consider the 4×4 matrix $M = \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix}$. Then the value of the determinant of M is equal to _________.
Q.58	Let σ be the permutation in the symmetric group S_5 given by $\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1, \quad \sigma(4) = 5, \quad \sigma(5) = 4.$ Define $N(\sigma) = \{\tau \in S_5 : \sigma \circ \tau = \tau \circ \sigma\}.$ Then the number of elements in $N(\sigma)$ is equal to __________.
Q.59	Let $f: (-1, 1) \to \mathbb{R}$ and $g: (-1, 1) \to \mathbb{R}$ be thrice continuously differentiable functions such that $f(x) \neq g(x)$ for every nonzero $x \in (-1, 1)$. Suppose $f(0) = \ln 2, \quad f'(0) = \pi, \quad f''(0) = \pi^2, \quad \text{and} \quad f'''(0) = \pi^9$ and $g(0) = \ln 2, \quad g'(0) = \pi, \quad g''(0) = \pi^2, \quad \text{and} \quad g'''(0) = \pi^3.$ Then the value of the limit $\lim_{x \to 0} \frac{e^{f(x)} - e^{g(x)}}{f(x) - g(x)}$ is equal to _________. (*Rounded off to two decimal places*)

Q.60	If $f: [0, \infty) \to \mathbb{R}$ and $g: [0, \infty) \to [0, \infty)$ are continuous functions such that $\int_0^{x^3+x^2} f(t)dt = x^2$ and $\int_0^{g(x)} t^2 dt = 9(x+1)^3$ for all $x \in [0, \infty)$, then the value of $$f(2) + g(2) + 16\, f(12)$$ is equal to __________. (*Rounded off to two decimal places*)

Question No.	Question Type (QT)	Subject Name (SN)	Key/Range (KY)	Mark (MK)
1	MCQ	MA	C	1
2	MCQ	MA	D	1
3	MCQ	MA	C	1
4	MCQ	MA	B	1
5	MCQ	MA	B	1
6	MCQ	MA	D	1
7	MCQ	MA	C	1
8	MCQ	MA	A	1
9	MCQ	MA	A	1
10	MCQ	MA	D	1
11	MCQ	MA	C	2
12	MCQ	MA	C	2
13	MCQ	MA	B	2
14	MCQ	MA	D	2
15	MCQ	MA	Marks to All	2
16	MCQ	MA	A	2
17	MCQ	MA	B	2
18	MCQ	MA	C	2
19	MCQ	MA	A	2
20	MCQ	MA	C	2
21	MCQ	MA	D	2
22	MCQ	MA	C	2
23	MCQ	MA	A	2

Question No.	Question Type (QT)	Subject Name (SN)	Key/Range (KY)	Mark (MK)
24	MCQ	MA	B	2
25	MCQ	MA	B	2
26	MCQ	MA	B	2
27	MCQ	MA	D	2
28	MCQ	MA	B	2
29	MCQ	MA	C	2
30	MCQ	MA	C	2
31	MSQ	MA	A,C	2
32	MSQ	MA	A,B	2
33	MSQ	MA	A,B,C	2
34	MSQ	MA	A,D	2
35	MSQ	MA	C,D	2
36	MSQ	MA	C,D	2
37	MSQ	MA	A,C,D	2
38	MSQ	MA	A,C	2
39	MSQ	MA	A,B	2
40	MSQ	MA	A,C	2
41	NAT	MA	2.19 to 2.21	1
42	NAT	MA	14 to 14	1
43	NAT	MA	0.32 to 0.34	1
44	NAT	MA	0.24 to 0.26	1
45	NAT	MA	8 to 8	1
46	NAT	MA	420 to 420	1

Question No.	Question Type (QT)	Subject Name (SN)	Key/Range (KY)	Mark (MK)
47	NAT	MA	0 to 0	1
48	NAT	MA	0.49 to 0.51	1
49	NAT	MA	2.24 to 2.26	1
50	NAT	MA	2.00 to 2.00	1
51	NAT	MA	18 to 18	2
52	NAT	MA	4.15 to 4.17	2
53	NAT	MA	3.00 to 3.00	2
54	NAT	MA	2.00 to 2.00	2
55	NAT	MA	3.00 to 3.00	2
56	NAT	MA	5 to 5	2
57	NAT	MA	41 to 41	2
58	NAT	MA	6 to 6	2
59	NAT	MA	2.00 to 2.00	2
60	NAT	MA	13.39 to 13.41	2

SECTION – A

MULTIPLE CHOICE QUESTIONS (MCQ)

Q. 1 – Q. 10 carry one mark each.

Q. 1 Let G be a finite group. Then G is necessarily a cyclic group if the order of G is

(A) 4

(B) 7

(C) 6

(D) 10

Q. 2 Let $\mathbf{v}_1, \ldots, \mathbf{v}_9$ be the column vectors of a non-zero 9×9 real matrix A. Let $a_1, \ldots, a_9 \in \mathbb{R}$, not all zero, be such that $\sum_{i=1}^{9} a_i\mathbf{v}_i = \mathbf{0}$. Then the system $A\mathbf{x} = \sum_{i=1}^{9} \mathbf{v}_i$ has

(A) no solution

(B) a unique solution

(C) more than one but only finitely many solutions

(D) infinitely many solutions

Q. 3 Which of the following is a subspace of the real vector space $\mathbb{R}^3$?

(A) $\{(x, y, z) \in \mathbb{R}^3 : (y + z)^2 + (2x - 3y)^2 = 0\}$

(B) $\{(x, y, z) \in \mathbb{R}^3 : y \in \mathbb{Q}\}$

(C) $\{(x, y, z) \in \mathbb{R}^3 : yz = 0\}$

(D) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z + 1 = 0\}$

Q. 4 Consider the initial value problem

$$\frac{dy}{dx} + \alpha y = 0,$$
$$y(0) = 1,$$

where $\alpha \in \mathbb{R}$. Then

(A) there is an α such that $y(1) = 0$

(B) there is a unique α such that $\lim_{x \to \infty} y(x) = 0$

(C) there is NO α such that $y(2) = 1$

(D) there is a unique α such that $y(1) = 2$

Q. 5 Let $p(x) = x^{57} + 3x^{10} - 21x^3 + x^2 + 21$ and

$$q(x) = p(x) + \sum_{j=1}^{57} p^{(j)}(x) \quad \text{for all } x \in \mathbb{R},$$

where $p^{(j)}(x)$ denotes the j^{th} derivative of $p(x)$. Then the function q admits

(A) NEITHER a global maximum NOR a global minimum on $\mathbb{R}$

(B) a global maximum but NOT a global minimum on $\mathbb{R}$

(C) a global minimum but NOT a global maximum on $\mathbb{R}$

(D) a global minimum and a global maximum on $\mathbb{R}$

Q. 6 The limit

$$\lim_{a \to 0} \left(\frac{\int_0^a \sin(x^2)\, dx}{\int_0^a (\ln(x+1))^2\, dx} \right)$$

is

(A) 0

(B) 1

(C) $\frac{\pi}{e}$

(D) non-existent

Q. 7 The value of

$$\int_0^1 \int_0^{1-x} \cos(x^3 + y^2)\, dy\, dx - \int_0^1 \int_0^{1-y} \cos(x^3 + y^2)\, dx\, dy$$

is

(A) 0

(B) $\dfrac{\cos(1)}{2}$

(C) $\dfrac{\sin(1)}{2}$

(D) $\cos\left(\dfrac{1}{2}\right) - \sin\left(\dfrac{1}{2}\right)$

Q. 8 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (e^x \cos(y), e^x \sin(y))$. Then the number of points in $\mathbb{R}^2$ that do NOT lie in the range of f is

(A) 0

(B) 1

(C) 2

(D) infinite

Q. 9 Let $a_n = \left(1 + \frac{1}{n}\right)^n$ and $b_n = n \cos\left(\frac{n!\pi}{2^{10}}\right)$ for $n \in \mathbb{N}$. Then

(A) (a_n) is convergent and (b_n) is bounded

(B) (a_n) is NOT convergent and (b_n) is bounded

(C) (a_n) is convergent and (b_n) is unbounded

(D) (a_n) is NOT convergent and (b_n) is unbounded

Q. 10 Let (a_n) be a sequence of real numbers defined by

$$a_n = \begin{cases} 1 & \text{if } n \text{ is prime} \\ -1 & \text{if } n \text{ is not prime.} \end{cases}$$

Let $b_n = \frac{a_n}{n}$ for $n \in \mathbb{N}$. Then

(A) both (a_n) and (b_n) are convergent

(B) (a_n) is convergent but (b_n) is NOT convergent

(C) (a_n) is NOT convergent but (b_n) is convergent

(D) both (a_n) and (b_n) are NOT convergent

Q. 11 – Q. 30 carry two marks each.

Q. 11 Let $a_n = \sin\left(\frac{1}{n^3}\right)$ and $b_n = \sin\left(\frac{1}{n}\right)$ for $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

(B) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is NOT convergent

(C) $\sum_{n=1}^{\infty} a_n$ is NOT convergent but $\sum_{n=1}^{\infty} b_n$ is convergent

(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are NOT convergent

Q. 12 Consider the following statements:

I. There exists a linear transformation from $\mathbb{R}^3$ to itself such that its range space and null space are the same.

II. There exists a linear transformation from $\mathbb{R}^2$ to itself such that its range space and null space are the same.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Q. 13 Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$$

and $B = A^5 + A^4 + I_3$. Which of the following is NOT an eigenvalue of B?

(A) 1

(B) 2

(C) 49

(D) 3

Q. 14 The system of linear equations in x_1, x_2, x_3

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{R}$, has

(A) at least one solution for any α and β

(B) a unique solution for any β when $\alpha \neq 1$

(C) NO solution for any α when $\beta \neq 5$

(D) infinitely many solutions for any α when $\beta = 5$

Q. 15 Let S and T be non-empty subsets of $\mathbb{R}^2$, and W be a non-zero proper subspace of $\mathbb{R}^2$. Consider the following statements:

I. If $\operatorname{span}(S) = \mathbb{R}^2$, then $\operatorname{span}(S \cap W) = W$.

II. $\operatorname{span}(S \cup T) = \operatorname{span}(S) \cup \operatorname{span}(T)$.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Q. 16 Let $f(x, y) = e^{x^2+y^2}$ for $(x, y) \in \mathbb{R}^2$, and a_n be the determinant of the matrix

$$\begin{pmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x \partial y} \\ \\ \dfrac{\partial^2 f}{\partial y \partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

evaluated at the point $(\cos(n), \sin(n))$. Then the limit $\lim\limits_{n\to\infty} a_n$ is

(A) non-existent

(B) 0

(C) $6e^2$

(D) $12e^2$

Q. 17 Let $f(x, y) = \ln(1 + x^2 + y^2)$ for $(x, y) \in \mathbb{R}^2$. Define

$$P = \left.\frac{\partial^2 f}{\partial x^2}\right|_{(0,0)} \qquad Q = \left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(0,0)}$$

$$R = \left.\frac{\partial^2 f}{\partial y \partial x}\right|_{(0,0)} \qquad S = \left.\frac{\partial^2 f}{\partial y^2}\right|_{(0,0)}$$

Then

(A) $PS - QR > 0$ and $P < 0$

(B) $PS - QR > 0$ and $P > 0$

(C) $PS - QR < 0$ and $P > 0$

(D) $PS - QR < 0$ and $P < 0$

Q. 18 The area of the curved surface

$$S = \{(x, y, z) \in \mathbb{R}^3 : z^2 = (x-1)^2 + (y-2)^2\}$$

lying between the planes $z = 2$ and $z = 3$ is

(A) $4\pi\sqrt{2}$

(B) $5\pi\sqrt{2}$

(C) 9π

(D) $9\pi\sqrt{2}$

Q. 19 Let $a_n = \dfrac{1 + 2^{-2} + \cdots + n^{-2}}{n}$ for $n \in \mathbb{N}$. Then

(A) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are convergent

(B) the sequence (a_n) is convergent but the series $\sum_{n=1}^{\infty} a_n$ is NOT convergent

(C) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are NOT convergent

(D) the sequence (a_n) is NOT convergent but the series $\sum_{n=1}^{\infty} a_n$ is convergent

Q. 20 Let (a_n) be a sequence of real numbers such that the series $\sum_{n=0}^{\infty} a_n(x-2)^n$ converges at $x = -5$. Then this series also converges at

(A) $x = 9$

(B) $x = 12$

(C) $x = 5$

(D) $x = -6$

Q. 21 Let (a_n) and (b_n) be sequences of real numbers such that

$$|a_n - a_{n+1}| = \frac{1}{2^n} \quad \text{and} \quad |b_n - b_{n+1}| = \frac{1}{\sqrt{n}} \quad \text{for } n \in \mathbb{N}.$$

Then

(A) both (a_n) and (b_n) are Cauchy sequences

(B) (a_n) is a Cauchy sequence but (b_n) need NOT be a Cauchy sequence

(C) (a_n) need NOT be a Cauchy sequence but (b_n) is a Cauchy sequence

(D) both (a_n) and (b_n) need NOT be Cauchy sequences

Q. 22 Consider the family of curves $x^2 + y^2 = 2x + 4y + k$ with a real parameter $k > -5$. Then the orthogonal trajectory to this family of curves passing through $(2, 3)$ also passes through

(A) $(3, 4)$

(B) $(-1, 1)$

(C) $(1, 0)$

(D) $(3, 5)$

Q. 23 Consider the following statements:

I. Every infinite group has infinitely many subgroups.

II. There are only finitely many non-isomorphic groups of a given finite order.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) I is FALSE but II is TRUE

(D) both I and II are FALSE

Q. 24 Suppose $f : (-1, 1) \to \mathbb{R}$ is an infinitely differentiable function such that the series $\sum_{j=0}^{\infty} a_j \frac{x^j}{j!}$ converges to $f(x)$ for each $x \in (-1, 1)$, where,

$$a_j = \int_0^{\pi/2} \theta^j \cos^j(\tan\theta)d\theta + \int_{\pi/2}^{\pi} (\theta - \pi)^j \cos^j(\tan\theta)d\theta$$

for $j \geq 0$. Then

(A) $f(x) = 0$ for all $x \in (-1, 1)$

(B) f is a non-constant even function on $(-1, 1)$

(C) f is a non-constant odd function on $(-1, 1)$

(D) f is NEITHER an odd function NOR an even function on $(-1, 1)$

Q. 25 Let $f(x) = \cos(x)$ and $g(x) = 1 - \dfrac{x^2}{2}$ for $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$. Then

(A) $f(x) \geq g(x)$ for all $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(B) $f(x) \leq g(x)$ for all $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(C) $f(x) - g(x)$ changes sign exactly once on $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(D) $f(x) - g(x)$ changes sign more than once on $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

Q. 26 Let

$$f(x, y) = \iint_{(u-x)^2+(v-y)^2 \leq 1} e^{-\sqrt{(u-x)^2+(v-y)^2}}\, du\, dv.$$

Then $\lim\limits_{n\to\infty} f(n, n^2)$ is

(A) non-existent

(B) 0

(C) $\pi(1 - e^{-1})$

(D) $2\pi(1 - 2e^{-1})$

Q. 27 How many group homomorphisms are there from $\mathbb{Z}_2$ to S_5?

(A) 40

(B) 41

(C) 26

(D) 25

Q. 28 Let $y : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that y'' is continuous on $[0, 1]$ and $y(0) = y(1) = 0$. Suppose $y''(x) + x^2 < 0$ for all $x \in [0, 1]$. Then

(A) $y(x) > 0$ for all $x \in (0, 1)$

(B) $y(x) < 0$ for all $x \in (0, 1)$

(C) $y(x) = 0$ has exactly one solution in $(0, 1)$

(D) $y(x) = 0$ has more than one solution in $(0, 1)$

Q. 29 From the additive group $\mathbb{Q}$ to which one of the following groups does there exist a non-trivial group homomorphism?

(A) $\mathbb{R}^\times$, the multiplicative group of non-zero real numbers

(B) $\mathbb{Z}$, the additive group of integers

(C) $\mathbb{Z}_2$, the additive group of integers modulo 2

(D) $\mathbb{Q}^\times$, the multiplicative group of non-zero rational numbers

Q. 30 Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that f'' has exactly two distinct zeroes. Then

(A) f' has at most 3 distinct zeroes

(B) f' has at least 1 zero

(C) f has at most 3 distinct zeroes

(D) f has at least 2 distinct zeroes

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SECTION – B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Q. 31 For each $t \in (0,1)$, the surface P_t in $\mathbb{R}^3$ is defined by

$$P_t = \{(x,y,z) : (x^2+y^2)z = 1,\ t^2 \le x^2+y^2 \le 1\}.$$

Let $a_t \in \mathbb{R}$ be the surface area of P_t. Then

(A) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^4} + \frac{4y^2}{(x^2+y^2)^4}}\, dx\, dy$

(B) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^2} + \frac{4y^2}{(x^2+y^2)^2}}\, dx\, dy$

(C) the limit $\lim_{t \to 0^+} a_t$ does NOT exist

(D) the limit $\lim_{t \to 0^+} a_t$ exists

Q. 32 Let $A \subseteq \mathbb{Z}$ with $0 \in A$. For $r, s \in \mathbb{Z}$, define

$$rA = \{ra : a \in A\}, \qquad rA + sA = \{ra + sb :\ a, b \in A\}.$$

Which of the following conditions imply that A is a subgroup of the additive group $\mathbb{Z}$?

(A) $-2A \subseteq A,\ \ A + A = A$

(B) $A = -A,\ \ A + 2A = A$

(C) $A = -A,\ \ A + A = A$

(D) $2A \subseteq A,\ \ A + A = A$

Q. 33 Let $y : (\sqrt{2/3}, \infty) \to \mathbb{R}$ be the solution of

$$(2x - y)y' + (2y - x) = 0,$$
$$y(1) = 3.$$

Then

(A) $y(3) = 1$

(B) $y(2) = 4 + \sqrt{10}$

(C) y' is bounded on $(\sqrt{2/3}, 1)$

(D) y' is bounded on $(1, \infty)$

Q. 34 Let $f : (-1, 1) \to \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$. Suppose there exists an $M > 0$ such that $|f'(x)| \leq M|x|$ for all $x \in (-1, 1)$. Then

(A) f' is continuous at $x = 0$

(B) f' is differentiable at $x = 0$

(C) ff' is differentiable at $x = 0$

(D) $(f')^2$ is differentiable at $x = 0$

Q. 35 Which of the following functions is/are Riemann integrable on $[0, 1]$?

(A) $f(x) = \int\limits_0^x \left|\frac{1}{2} - t\right| dt$

(B) $f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(C) $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0, 1] \\ -1 & \text{otherwise} \end{cases}$

(D) $f(x) = \begin{cases} x & \text{if } x \in [0, 1) \\ 0 & \text{if } x = 1 \end{cases}$

Q. 36 A subset $S \subseteq \mathbb{R}^2$ is said to be *bounded* if there is an $M > 0$ such that $|x| \leq M$ and $|y| \leq M$ for all $(x, y) \in S$. Which of the following subsets of $\mathbb{R}^2$ is/are bounded?

(A) $\{(x, y) \in \mathbb{R}^2 : \; e^{x^2} + y^2 \leq 4\}$

(B) $\{(x, y) \in \mathbb{R}^2 : \; x^4 + y^2 \leq 4\}$

(C) $\{(x, y) \in \mathbb{R}^2 : \; |x| + |y| \leq 4\}$

(D) $\{(x, y) \in \mathbb{R}^2 : \; e^{x^3} + y^2 \leq 4\}$

Q. 37 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as follows:

$$f(x,y) = \begin{cases} \dfrac{x^4y^3}{x^6+y^6} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Then

(A) $\displaystyle\lim_{t\to 0} \frac{f(t,t)-f(0,0)}{t}$ exists and equals $\dfrac{1}{2}$

(B) $\left.\dfrac{\partial f}{\partial x}\right|_{(0,0)}$ exists and equals 0

(C) $\left.\dfrac{\partial f}{\partial y}\right|_{(0,0)}$ exists and equals 0

(D) $\displaystyle\lim_{t\to 0} \frac{f(t,2t)-f(0,0)}{t}$ exists and equals $\dfrac{1}{3}$

Q. 38 Which of the following is/are true?

(A) Every linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto points or lines

(B) Every surjective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto lines

(C) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of parallel lines to pairs of parallel lines

(D) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of perpendicular lines to pairs of perpendicular lines

Q. 39 Which of the following is/are linear transformations?

(A) $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = \sin(x)$

(B) $T : M_2(\mathbb{R}) \to \mathbb{R}$ given by $T(A) = \text{trace}(A)$

(C) $T : \mathbb{R}^2 \to \mathbb{R}$ given by $T(x, y) = x + y + 1$

(D) $T : P_2(\mathbb{R}) \to \mathbb{R}$ given by $T(p(x)) = p(1)$

Q. 40 Let R_1 and R_2 be the radii of convergence of the power series $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ and $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$, respectively. Then

(A) $R_1 = R_2$

(B) $R_2 > 1$

(C) $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ converges for all $x \in [-1, 1]$

(D) $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$ converges for all $x \in [-1, 1]$

SECTION – C

NUMERICAL ANSWER TYPE (NAT)

Q. 41 – Q. 50 carry one mark each.

Q. 41 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined as follows:

$$f(x,y) = \begin{cases} (x^2-1)^2 \cos^2\left(\dfrac{y^2}{(x^2-1)^2}\right) & \text{if } x \neq \pm 1 \\ 0 & \text{if } x = \pm 1. \end{cases}$$

The number of points of discontinuity of $f(x,y)$ is equal to ____________.

Q. 42 Let $T : P_2(\mathbb{R}) \to P_4(\mathbb{R})$ be the linear transformation given by $T(p(x)) = p(x^2)$. Then the rank of T is equal to ____________.

Q. 43 If y is the solution of

$$y'' - 2y' + y = e^x,$$

$$y(0) = 0, \quad y'(0) = -1/2,$$

then $y(1)$ is equal to ____________. (rounded off to two decimal places)

Q. 44 The value of

$$\lim_{n\to\infty}\left(n\int_0^1 \frac{x^n}{x+1}\,dx\right)$$

is equal to ____________. (rounded off to two decimal places)

Q. 45 For $\sigma \in S_8$, let $o(\sigma)$ denote the order of σ. Then $\max\{o(\sigma) : \sigma \in S_8\}$ is equal to ____________.

Q. 46 For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_8$ denote the residue class of g modulo 8. Consider the group $\mathbb{Z}_8^\times = \{\bar{x} \in \mathbb{Z}_8 : 1 \leq x \leq 7, \gcd(x,8) = 1\}$ with respect to multiplication modulo 8. The number of group isomorphisms from $\mathbb{Z}_8^\times$ onto itself is equal to ____________.

Q. 47 Let $f(x) = \sqrt[3]{x}$ for $x \in (0, \infty)$, and $\theta(h)$ be a function such that

$$f(3+h) - f(3) = hf'(3 + \theta(h)h)$$

for all $h \in (-1, 1)$. Then $\lim_{h \to 0} \theta(h)$ is equal to ____________.
(rounded off to two decimal places)

Q. 48 Let V be the volume of the region $S \subseteq \mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : \ xy \leq z \leq 4, \ 0 \leq x^2 + y^2 \leq 1\}.$$

Then $\dfrac{V}{\pi}$ is equal to ____________. (rounded off to two decimal places)

Q. 49 The sum of the series $\displaystyle\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}$ is equal to ____________.
(rounded off to two decimal places)

Q. 50 The value of $\displaystyle\lim_{n \to \infty} \left(1 + \frac{1}{2^n} + \frac{1}{3^n} + \cdots + \frac{1}{(2023)^n}\right)^{\frac{1}{n}}$ is equal to ____________.
(rounded off to two decimal places)

Q. 51 – Q. 60 carry two marks each.

Q. 51 Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined as $f(x, y, z) = x^3 + y^3 + z^3$, and let $L : \mathbb{R}^3 \to \mathbb{R}$ be the linear map satisfying

$$\lim_{(x,y,z)\to(0,0,0)} \frac{f(1 + x, 1 + y, 1 + z) - f(1, 1, 1) - L(x, y, z)}{\sqrt{x^2 + y^2 + z^2}} = 0.$$

Then $L(1, 2, 4)$ is equal to ____________. (rounded off to two decimal places)

Q. 52 The global minimum value of

$$f(x) = |x - 1| + |x - 2|^2$$

on $\mathbb{R}$ is equal to ____________. (rounded off to two decimal places)

Q. 53 Let $y : (1, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$y'' - \frac{2y}{(1 - x)^2} = 0$$

satisfying $y(2) = 1$ and $\lim_{x\to\infty} y(x) = 0$. Then $y(3)$ is equal to ____________. (rounded off to two decimal places)

Q. 54 The number of permutations in S_4 that have exactly two cycles in their cycle decompositions is equal to ____________.

Q. 55 Let S be the triangular region whose vertices are $(0, 0)$, $\left(0, \frac{\pi}{2}\right)$, and $\left(\frac{\pi}{2}, 0\right)$. The value of $\iint_S \sin(x)\cos(y)\, dx\, dy$ is equal to ____________. (rounded off to two decimal places)

Q. 56 Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the maximum possible rank of B is equal to ____________.

Q. 57 Let W be the subspace of $M_3(\mathbb{R})$ consisting of all matrices with the property that the sum of the entries in each row is zero and the sum of the entries in each column is zero. Then the dimension of W is equal to ____________.

Q. 58 The maximum number of linearly independent eigenvectors of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is equal to ____________.

Q. 59 Let S be the set of all real numbers α such that the solution y of the initial value problem

$$\frac{dy}{dx} = y(2 - y),$$
$$y(0) = \alpha,$$

exists on $[0, \infty)$. Then the minimum of the set S is equal to ____________. (rounded off to two decimal places)

Q. 60 Let $f : \mathbb{R} \to \mathbb{R}$ be a bijective function such that for all $x \in \mathbb{R}$, $f(x) = \sum_{n=1}^{\infty} a_n x^n$ and $f^{-1}(x) = \sum_{n=1}^{\infty} b_n x^n$, where f^{-1} is the inverse function of f. If $a_1 = 2$ and $a_2 = 4$, then b_1 is equal to ____________.

Answer Key

MA : JAM 2023

Question No.	Question ID	Question Type	Section	Answer	Marks
1	3651212521	MCQ	A	B	1
2	3651212522	MCQ	A	D	1
3	3651212523	MCQ	A	A	1
4	3651212524	MCQ	A	D	1
5	3651212525	MCQ	A	A	1
6	3651212526	MCQ	A	B	1
7	3651212527	MCQ	A	A	1
8	3651212528	MCQ	A	B	1
9	3651212529	MCQ	A	C	1
10	3651212530	MCQ	A	C	1
11	3651212531	MCQ	A	B	2
12	3651212532	MCQ	A	C	2
13	3651212533	MCQ	A	B	2
14	3651212534	MCQ	A	B	2
15	3651212535	MCQ	A	D	2
16	3651212536	MCQ	A	D	2
17	3651212537	MCQ	A	B	2
18	3651212538	MCQ	A	B	2
19	3651212539	MCQ	A	B	2
20	3651212540	MCQ	A	C	2
21	3651212541	MCQ	A	B	2
22	3651212542	MCQ	A	A	2
23	3651212543	MCQ	A	A	2
24	3651212544	MCQ	A	B	2
25	3651212545	MCQ	A	A	2
26	3651212546	MCQ	A	D	2
27	3651212547	MCQ	A	C	2
28	3651212548	MCQ	A	A	2
29	3651212549	MCQ	A	A	2
30	3651212550	MCQ	A	A	2
31	3651212551	MSQ	B	A, C	2
32	3651212552	MSQ	B	A, B, C	2
33	3651212553	MSQ	B	B, D	2
34	3651212554	MSQ	B	A, C, D	2
35	3651212555	MSQ	B	A, B, D	2
36	3651212556	MSQ	B	A, B, C	2
37	3651212557	MSQ	B	A, B, C	2
38	3651212558	MSQ	B	A, B, C	2
39	3651212559	MSQ	B	B, D	2
40	3651212560	MSQ	B	A, D	2
41	3651212561	NAT	C	0 to 0	1
42	3651212562	NAT	C	3 to 3	1
43	3651212563	NAT	C	-0.01 to 0.01	1
44	3651212564	NAT	C	0.49 to 0.51	1
45	3651212565	NAT	C	15 to 15	1
46	3651212566	NAT	C	6 to 6	1
47	3651212567	NAT	C	0.49 to 0.51	1
48	3651212568	NAT	C	3.99 to 4.01	1
49	3651212569	NAT	C	0.49 to 0.51	1
50	3651212570	NAT	C	0.99 to 1.01	1
51	3651212571	NAT	C	20.99 to 21.01	2
52	3651212572	NAT	C	0.74 to 0.76	2
53	3651212573	NAT	C	0.49 to 0.51	2
54	3651212574	NAT	C	11 to 11	2
55	3651212575	NAT	C	0.49 to 0.51	2
56	3651212576	NAT	C	2 to 2	2
57	3651212577	NAT	C	4 to 4	2
58	3651212578	NAT	C	3 to 3	2
59	3651212579	NAT	C	-0.01 to 0.01	2
60	3651212580	NAT	C	0.49 to 0.51	2

Special Instructions / Useful Data

$\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$ denotes the additive group of integers modulo n

$\mathbb{R} =$ the set of all real numbers

$\mathbb{N} =$ the set of all positive integers

$\mathbb{Z} =$ the set of all integers

$\mathbb{C} =$ the set of all complex numbers

$\mathbb{Q} =$ the set of all rational numbers

$\gcd(r, n) =$ the greatest common divisor of the integers r and n

$S_n =$ the symmetric group of all permutations of $\{1,2, \dots, n\}$

A_n = the group of all even permutations in S_n

$M_n(\mathbb{C}) =$ the set of all $n \times n$ matrices with entries from $\mathbb{C}$

$M_n(\mathbb{R}) =$ the set of all $n \times n$ matrices with entries from $\mathbb{R}$

$M^T =$ the transpose of the matrix M

$I_n =$ the $n \times n$ identity matrix

$P_n(x) =$ the real vector space of polynomials, in the variable x with real coefficients and having degree at most n, together with the zero polynomial. These polynomials are regarded as functions from $\mathbb{R}$ to $\mathbb{R}$

$\binom{n}{k} =$ the binomial coefficient defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

$f \circ g =$ the composite function defined by $(f \circ g)(x) = f(g(x))$

$A \setminus B$ = the complement of the set B in the set A, that is, $\{x \in A : x \notin B\}$

$\log x =$ the logarithm of x to the base e for a positive number x

$\mathbb{R}^n =$ the n-dimensional Euclidean space

$A \times B$ = the Cartesian product of the sets A and B

$M^{-1} =$ the inverse of an invertible matrix M

Section A: Q.1 – Q.10 Carry ONE mark each.

Q.1 Let $y_c: \mathbb{R} \to (0, \infty)$ be the solution of the Bernoulli's equation

$$\frac{dy}{dx} - y + y^3 = 0, \qquad y(0) = c > 0.$$

Then, for every $c > 0$, which one of the following is true?

(A) $\lim_{x\to\infty} y_c(x) = 0$

(B) $\lim_{x\to\infty} y_c(x) = 1$

(C) $\lim_{x\to\infty} y_c(x) = e$

(D) $\lim_{x\to\infty} y_c(x)$ does not exist

Q.2 For a twice continuously differentiable function $g: \mathbb{R} \to \mathbb{R}$, define

$$u_g(x,y) = \frac{1}{y}\int_{-y}^{y} g(x+t)dt \quad \text{for } (x,y) \in \mathbb{R}^2, \qquad y > 0.$$

Which one of the following holds for all such g?

(A) $\dfrac{\partial^2 u_g}{\partial x^2} = \dfrac{2}{y}\dfrac{\partial u_g}{\partial y} + \dfrac{\partial^2 u_g}{\partial y^2}$

(B) $\dfrac{\partial^2 u_g}{\partial x^2} = \dfrac{1}{y}\dfrac{\partial u_g}{\partial y} + \dfrac{\partial^2 u_g}{\partial y^2}$

(C) $\dfrac{\partial^2 u_g}{\partial x^2} = \dfrac{2}{y}\dfrac{\partial u_g}{\partial y} - \dfrac{\partial^2 u_g}{\partial y^2}$

(D) $\dfrac{\partial^2 u_g}{\partial x^2} = \dfrac{1}{y}\dfrac{\partial u_g}{\partial y} - \dfrac{\partial^2 u_g}{\partial y^2}$

Q.3 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = 1 + y \sec x \quad \text{for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

that satisfies $y(0) = 0$. Then, the value of $y\left(\frac{\pi}{6}\right)$ equals

(A) $\sqrt{3}\ \log\left(\frac{3}{2}\right)$

(B) $\left(\frac{\sqrt{3}}{2}\right)\log\left(\frac{3}{2}\right)$

(C) $\left(\frac{\sqrt{3}}{2}\right)\log 3$

(D) $\sqrt{3}\ \log 3$

Q.4 Let $\mathcal{F}$ be the family of curves given by

$$x^2 + 2hxy + y^2 = 1, \quad -1 < h < 1\,.$$

Then, the differential equation for the family of orthogonal trajectories to $\mathcal{F}$ is

(A) $(x^2y - y^3 + y)\dfrac{dy}{dx} - (xy^2 - x^3 + x) = 0$

(B) $(x^2y - y^3 + y)\dfrac{dy}{dx} + (xy^2 - x^3 + x) = 0$

(C) $(x^2y + y^3 + y)\dfrac{dy}{dx} - (xy^2 + x^3 + x) = 0$

(D) $(x^2y + y^3 + y)\dfrac{dy}{dx} + (xy^2 + x^3 + x) = 0$

Q.5 Let G be a group of order 39 such that it has exactly one subgroup of order 3 and exactly one subgroup of order 13. Then, which one of the following statements is TRUE?

(A) G is necessarily cyclic

(B) G is abelian but need not be cyclic

(C) G need not be abelian

(D) G has 13 elements of order 13

Q.6 For a positive integer n, let $U(n) = \{\bar{r} \in \mathbb{Z}_n \, : \, \gcd(r, n) = 1\}$ be the group under multiplication modulo n. Then, which one of the following statements is TRUE?

(A) $U(5)$ is isomorphic to $U(8)$

(B) $U(10)$ is isomorphic to $U(12)$

(C) $U(8)$ is isomorphic to $U(10)$

(D) $U(8)$ is isomorphic to $U(12)$

Q.7 Which one of the following is TRUE for the symmetric group S_{13}?

(A) S_{13} has an element of order 42

(B) S_{13} has no element of order 35

(C) S_{13} has an element of order 27

(D) S_{13} has no element of order 60

Q.8 Let G be a finite group containing a non-identity element which is conjugate to its inverse. Then, which one of the following is TRUE?

(A) The order of G is necessarily even

(B) The order of G is not necessarily even

(C) G is necessarily cyclic

(D) G is necessarily abelian but need not be cyclic

Q.9 Consider the following statements.

P: If a system of linear equations $Ax = b$ has a unique solution, where A is an $m \times n$ matrix and b is an $m \times 1$ matrix, then $m = n$.

Q: For a subspace W of a nonzero vector space V, whenever $u \in V \setminus W$ and $v \in V \setminus W$, then $u + v \in V \setminus W$.

Which one of the following holds?

(A) Both P and Q are true

(B) P is true but Q is false

(C) P is false but Q is true

(D) Both P and Q are false

Q.10 Let $g: \mathbb{R} \to \mathbb{R}$ be a continuous function. Which one of the following is the solution of the differential equation

$$\frac{d^2y}{dx^2} + y = g(x) \quad \text{for} \quad x \in \mathbb{R},$$

satisfying the conditions $y(0) = 0,\ y'(0) = 1$?

(A) $y(x) = \sin x - \int_0^x \sin(x-t)\, g(t) dt$

(B) $y(x) = \sin x + \int_0^x \sin(x-t)\, g(t) dt$

(C) $y(x) = \sin x - \int_0^x \cos(x-t)\, g(t) dt$

(D) $y(x) = \sin x + \int_0^x \cos(x-t)\, g(t) dt$

Section A: Q.11 – Q.30 Carry TWO marks each.

Q.11 Which one of the following groups has elements of order $1, 2, 3, 4, 5$ but does not have an element of order greater than or equal to 6 ?

(A) The alternating group A_6

(B) The alternating group A_5

(C) S_6

(D) S_5

Q.12 Consider the group $G = \{A \in M_2(\mathbb{R}) : AA^T = I_2\}$ with respect to matrix multiplication. Let

$$Z(G) = \{A \in G : AB = BA, \text{ for all } \quad B \in G\}.$$

Then, the cardinality of $Z(G)$ is

(A) 1

(B) 2

(C) 4

(D) Infinite

Q.13 Let V be a nonzero subspace of the complex vector space $M_7(\mathbb{C})$ such that every nonzero matrix in V is invertible. Then, the dimension of V over $\mathbb{C}$ is

(A) 1

(B) 2

(C) 7

(D) 49

Q.14 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{(3n+2)(3n+4)} \quad \text{and} \quad b_n = \frac{n^3 + \cos(3^n)}{3^n + n^3}.$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is divergent

(B) $\sum_{n=1}^{\infty} a_n$ is divergent but $\sum_{n=1}^{\infty} b_n$ is convergent

(C) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent

(D) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

Q.15 Let $a = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \\ 0 \end{bmatrix}$. Consider the following two statements.

P: The matrix $I_4 - aa^T$ is invertible.

Q: The matrix $I_4 - 2aa^T$ is invertible.

Then, which one of the following holds?

(A) P is false but Q is true

(B) P is true but Q is false

(C) Both P and Q are true

(D) Both P and Q are false

Q.16 Let A be a 6×5 matrix with entries in $\mathbb{R}$ and B be a 5×4 matrix with entries in $\mathbb{R}$. Consider the following two statements.

P: For all such nonzero matrices A and B, there is a nonzero matrix Z such that AZB is the 6×4 zero matrix.

Q: For all such nonzero matrices A and B, there is a nonzero matrix Y such that BYA is the 5×5 zero matrix.

Which one of the following holds?

(A) Both P and Q are true

(B) P is true but Q is false

(C) P is false but Q is true

(D) Both P and Q are false

Q.17 Let $P_{11}(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 11, together with the zero polynomial. Let

$$E = \{s_0(x), s_1(x), \dots, s_{11}(x)\}, \qquad F = \{r_0(x), r_1(x), \dots, r_{11}(x)\}$$

be subsets of $P_{11}(x)$ having 12 elements each and satisfying

$$s_0(3) = s_1(3) = \cdots = s_{11}(3) = 0\,, \qquad r_0(4) = r_1(4) = \cdots = r_{11}(4) = 1\,.$$

Then, which one of the following is TRUE?

(A) Any such E is not necessarily linearly dependent and any such F is not necessarily linearly dependent

(B) Any such E is necessarily linearly dependent but any such F is not necessarily linearly dependent

(C) Any such E is not necessarily linearly dependent but any such F is necessarily linearly dependent

(D) Any such E is necessarily linearly dependent and any such F is necessarily linearly dependent

Q.18 For the differential equation

$$y(8x - 9y)dx + 2x(x - 3y)dy = 0\,,$$

which one of the following statements is TRUE?

(A) The differential equation is not exact and has x^2 as an integrating factor

(B) The differential equation is exact and homogeneous

(C) The differential equation is not exact and does not have x^2 as an integrating factor

(D) The differential equation is not homogeneous and has x^2 as an integrating factor

Q.19 For $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x.

For $x, y \in \mathbb{R}$, define

$$\min\{x, y\} = \begin{cases} x & \text{if } x \leq y, \\ y & \text{otherwise.} \end{cases}$$

Let $f: [-2\pi, 2\pi] \to \mathbb{R}$ be defined by

$$f(x) = \sin(\min\{x, x - \lfloor x \rfloor\}) \text{ for } x \in [-2\pi, 2\pi].$$

Consider the set $S = \{x \in [-2\pi, 2\pi]: f \text{ is discontinuous at } x\}$.

Which one of the following statements is TRUE?

(A) S has 13 elements

(B) S has 7 elements

(C) S is an infinite set

(D) S has 6 elements

Q.20 Define the sequences $\{a_n\}_{n=3}^{\infty}$ and $\{b_n\}_{n=3}^{\infty}$ as

$$a_n = (\log n + \log\log n)^{\log n} \quad \text{and} \quad b_n = n^{\left(1+\frac{1}{\log n}\right)}.$$

Which one of the following is TRUE?

(A) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is convergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is divergent

(B) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is divergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is convergent

(C) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are divergent

(D) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are convergent

Q.21 For $p, q, r \in \mathbb{R},\ r \neq 0$ and $n \in \mathbb{N}$, let

$$a_n = p^n\, n^q \left(\frac{n}{n+2}\right)^{n^2} \quad \text{and} \quad b_n = \frac{n^n}{n!\ r^n}\left(\sqrt{\frac{n+2}{n}}\right).$$

Then, which one of the following statements is TRUE?

(A) If $1 < p < e^2$ and $q > 1,$ then $\sum_{n=1}^{\infty} a_n$ is convergent

(B) If $\mathrm{e}^2 < p < e^4$ and $q > 1,$ then $\sum_{n=1}^{\infty} a_n$ is convergent

(C) If $1 < r < e,$ then $\sum_{n=1}^{\infty} b_n$ is convergent

(D) If $\frac{1}{\mathrm{e}} < r < 1,$ then $\sum_{n=1}^{\infty} b_n$ is convergent

Q.22 Let $P_7(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 7, together with the zero polynomial. Let $T\colon\ P_7(x) \to P_7(x)$ be the linear transformation defined by

$$T\big(f(x)\big) = f(x) + \frac{df(x)}{dx} .$$

Then, which one of the following is TRUE?

(A) T is not a surjective linear transformation

(B) There exists $k \in \mathbb{N}$ such that T^k is the zero linear transformation

(C) 1 and 2 are the eigenvalues of T

(D) There exists $r \in \mathbb{N}$ such that $(T - I)^r$ is the zero linear transformation, where I is the identity map on $P_7(x)$

Q.23 For $\alpha \in \mathbb{R}$, let $y_\alpha(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 2y = \frac{1}{1+x^2} \quad \text{for } x \in \mathbb{R}$$

satisfying $y(0) = \alpha$. Then, which one of the following is TRUE?

(A) $\lim_{x\to\infty} y_\alpha(x) = 0$ for every $\alpha \in \mathbb{R}$

(B) $\lim_{x\to\infty} y_\alpha(x) = 1$ for every $\alpha \in \mathbb{R}$

(C) There exists an $\alpha \in \mathbb{R}$ such that $\lim_{x\to\infty} y_\alpha(x)$ exists but its value is different from 0 and 1

(D) There is an $\alpha \in \mathbb{R}$ for which $\lim_{x\to\infty} y_\alpha(x)$ does not exist

Q.24 Consider the following two statements.

P: There exist functions $f: \mathbb{R} \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ such that f is continuous at $x = 1$ and g is discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Q: There exist functions $f: \mathbb{R} \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ such that both f and g are discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Which one of the following holds?

(A) Both P and Q are true

(B) Both P and Q are false

(C) P is true but Q is false

(D) P is false but Q is true

Q.25 Let $f: \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \frac{(x^2+1)^2}{x^4+x^2+1} \quad \text{for } x \in \mathbb{R}.$$

Then, which one of the following is TRUE?

(A) f has exactly two points of local maxima and exactly three points of local minima

(B) f has exactly three points of local maxima and exactly two points of local minima

(C) f has exactly one point of local maximum and exactly two points of local minima

(D) f has exactly two points of local maxima and exactly one point of local minimum

Q.26 Let $f: \mathbb{R} \to \mathbb{R}$ be a solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 2e^x \text{ for } x \in \mathbb{R}.$$

Consider the following statements.

P: If $f(x) > 0$ for all $x \in \mathbb{R}$, then $f'(x) > 0$ for all $x \in \mathbb{R}$.

Q: If $f'(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) > 0$ for all $x \in \mathbb{R}$.

Then, which one of the following holds?

(A) P is true but Q is false

(B) P is false but Q is true

(C) Both P and Q are true

(D) Both P and Q are false

Q.27 For $a > b > 0$, consider

$$D = \left\{(x, y, z) \in \mathbb{R}^3 :\ x^2 + y^2 + z^2 \leq a^2 \ \text{ and } \ x^2 + y^2 \geq b^2\right\}.$$

Then, the surface area of the boundary of the solid D is

(A) $4\pi(a + b)\sqrt{a^2 - b^2}$

(B) $4\pi\left(a^2 - b\sqrt{a^2 - b^2}\right)$

(C) $4\pi(a - b)\sqrt{a^2 - b^2}$

(D) $4\pi\left(a^2 + b\sqrt{a^2 - b^2}\right)$

Q.28 For $n \geq 3$, let a regular n-sided polygon P_n be circumscribed by a circle of radius R_n and let r_n be the radius of the circle inscribed in P_n. Then

$$\lim_{n\to\infty}\left(\frac{R_n}{r_n}\right)^{n^2}$$

equals

(A) $e^{\left(\pi^2\right)}$

(B) $e^{\left(\frac{\pi^2}{2}\right)}$

(C) $e^{\left(\frac{\pi^2}{3}\right)}$

(D) $e^{\left(2\pi^2\right)}$

Q.29 Let L_1 denote the line $y = 3x + 2$ and L_2 denote the line $y = 4x + 3$. Suppose that $f: \mathbb{R} \to \mathbb{R}$ is a four times continuously differentiable function such that the line L_1 intersects the curve $y = f(x)$ at exactly three distinct points and the line L_2 intersects the curve $y = f(x)$ at exactly four distinct points. Then, which one of the following is TRUE?

(A) $\frac{df}{dx}$ does not attain the value 3 on $\mathbb{R}$

(B) $\frac{d^2f}{dx^2}$ vanishes at most once on $\mathbb{R}$

(C) $\frac{d^3f}{dx^3}$ vanishes at least once on $\mathbb{R}$

(D) $\frac{df}{dx}$ does not attain the value $\frac{7}{2}$ on $\mathbb{R}$

Q.30 Define the function $f: \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x, y) = 12xy\, e^{-(2x+3y-2)} .$$

If (a, b) is the point of local maximum of f, then $f(a, b)$ equals

(A) 2

(B) 6

(C) 12

(D) 0

Section B: Q.31 – Q.40 Carry TWO marks each.

Q.31 Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers.

Then, which of the following statements is/are always TRUE?

(A) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^2$ converges absolutely

(B) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^3$ converges absolutely

(C) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges

(D) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^3$ converges

Q.32 Which of the following statements is/are TRUE?

(A) $\sum_{n=1}^{\infty} n \log\left(1+\frac{1}{n^3}\right)$ is convergent

(B) $\sum_{n=1}^{\infty} \left(1-\cos\left(\frac{1}{n}\right)\right) \log n$ is convergent

(C) $\sum_{n=1}^{\infty} n^2 \log\left(1+\frac{1}{n^3}\right)$ is convergent

(D) $\sum_{n=1}^{\infty} \left(1-\cos\left(\frac{1}{\sqrt{n}}\right)\right) \log n$ is convergent

Q.33 Which of the following statements is/are TRUE?

(A) The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers

(B) The multiplicative group of nonzero real numbers is isomorphic to the multiplicative group of nonzero complex numbers

(C) The additive group of real numbers is isomorphic to the multiplicative group of nonzero complex numbers

(D) The additive group of real numbers is isomorphic to the additive group of rational numbers

Q.34 Let $f: (1, \infty) \to (0, \infty)$ be a continuous function such that for every $n \in \mathbb{N}$, $f(n)$ is the smallest prime factor of n. Then, which of the following options is/are CORRECT?

(A) $\lim_{x \to \infty} f(x)$ exists

(B) $\lim_{x \to \infty} f(x)$ does not exist

(C) The set of solutions to the equation $f(x) = 2024$ is finite

(D) The set of solutions to the equation $f(x) = 2024$ is infinite

Q.35 Let

$$S = \{(x, y) \in \mathbb{R}^2 :\ x > 0, y > 0\},$$

and $f: S \to \mathbb{R}$ be given by

$$f(x, y) = 2x^2 + 3y^2 - \log x - \frac{1}{6}\log y.$$

Then, which of the following statements is/are TRUE?

(A) There is a unique point in S at which $f(x, y)$ attains a local maximum

(B) There is a unique point in S at which $f(x, y)$ attains a local minimum

(C) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is bounded

(D) For each point $(x_0, y_0) \in S$, the set $\{(x, y) \in S: f(x, y) = f(x_0, y_0)\}$ is unbounded

Q.36 The center $Z(G)$ of a group G is defined as

$$Z(G) = \{x \in G : xg = gx \text{ for all } g \in G\}.$$

Let $|G|$ denote the order of G. Then, which of the following statements is/are TRUE for any group G?

(A) If G is non-abelian and $Z(G)$ contains more than one element, then the center of the quotient group $G/Z(G)$ contains only one element

(B) If $|G| \geq 2$, then there exists a non-trivial homomorphism from $\mathbb{Z}$ to G

(C) If $|G| \geq 2$ and G is non-abelian, then there exists a non-identity isomorphism from G to itself

(D) If $|G| = p^3$, where p is a prime number, then G is necessarily abelian

Q.37 For a matrix M, let Rowspace(M) denote the linear span of the rows of M and Colspace(M) denote the linear span of the columns of M. Which of the following hold(s) for all $A, B, C \in M_{10}(\mathbb{R})$ satisfying $A = BC$?

(A) Rowspace(A) $\subseteq$ Rowspace(B)

(B) Rowspace(A) $\subseteq$ Rowspace(C)

(C) Colspace(A) $\subseteq$ Colspace(B)

(D) Colspace(A) $\subseteq$ Colspace(C)

Q.38 Define $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ as follows

$$f(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}\ (m!)^2} \text{ and } g(x) = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}\ (m+1)!\ m!} \quad \text{for } x \in \mathbb{R}.$$

Let $x_1, x_2, x_3, x_4 \in \mathbb{R}$ be such that $0 < x_1 < x_2$, $0 < x_3 < x_4$,

$$f(x_1) = f(x_2) = 0, \qquad f(x) \neq 0 \text{ when } x_1 < x < x_2,$$

$$g(x_3) = g(x_4) = 0 \quad \text{and } g(x) \neq 0 \text{ when } x_3 < x < x_4.$$

Then, which of the following statements is/are TRUE?

(A) The function f does not vanish anywhere in the interval (x_3, x_4)

(B) The function f vanishes exactly once in the interval (x_3, x_4)

(C) The function g does not vanish anywhere in the interval (x_1, x_2)

(D) The function g vanishes exactly once in the interval (x_1, x_2)

Q.39 For $0 < \alpha < 4$, define the sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers as follows:

$$x_1 = \alpha \quad \text{and} \quad x_{n+1} + 2 = -x_n(x_n - 4) \quad \text{for } n \in \mathbb{N}.$$

Which of the following is/are TRUE?

(A) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (0,1)$

(B) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (1,2)$

(C) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (2,3)$

(D) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (3,4)$

Q.40 Consider

$$G = \{m + n\sqrt{2} : m, n \in \mathbb{Z}\}$$

as a subgroup of the additive group $\mathbb{R}$.

Which of the following statements is/are TRUE?

(A) G is a cyclic subgroup of $\mathbb{R}$ under addition

(B) $G \cap I$ is non-empty for every non-empty open interval $I \subseteq \mathbb{R}$

(C) G is a closed subset of $\mathbb{R}$

(D) G is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$, where the group operation in $\mathbb{Z} \times \mathbb{Z}$ is defined by $(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$

Section C: Q.41 – Q.50 Carry ONE mark each.

Q.41 The area of the region

$$R = \left\{(x, y) \in \mathbb{R}^2 \colon\ 0 \leq x \leq 1, 0 \leq y \leq 1 \ \text{and}\ \frac{1}{4} \leq xy \leq \frac{1}{2}\right\}$$

is __________ (*rounded off to two decimal places*).

Q.42 Let $y \colon \mathbb{R} \to \mathbb{R}$ be the solution to the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 1$$

satisfying $y(0) = 0$ and $y'(0) = 1$.

Then, $\lim_{x\to\infty} y(x)$ equals __________ (*rounded off to two decimal places*).

Q.43 For $\alpha > 0$, let $y_\alpha(x)$ be the solution to the differential equation

$$2\frac{d^2y}{dx^2} - \frac{dy}{dx} - y = 0$$

satisfying the conditions

$$y(0) = 1, \quad y'(0) = \alpha.$$

Then, the smallest value of α for which $y_\alpha(x)$ has no critical points in $\mathbb{R}$ equals __________ (*rounded off to the nearest integer*).

Q.44 Consider the 4×4 matrix

$$M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}.$$

If $a_{i,j}$ denotes the $(i,j)^{\text{th}}$ entry of M^{-1}, then $a_{4,1}$ equals __________ (*rounded off to two decimal places*).

Q.45 Let $P_{12}(x)$ be the real vector space of polynomials in the variable x with real coefficients and having degree at most 12, together with the zero polynomial. Define

$$V = \left\{ f \in P_{12}(x) \colon \ f(-x) = f(x) \text{ for all } x \in \mathbb{R} \text{ and } f(2024) = 0 \right\}.$$

Then, the dimension of V is ______________

Q.46 Let

$$S = \left\{ f \colon \mathbb{R} \to \mathbb{R} : \ f \text{ is a polynomial and } f\big(f(x)\big) = \big(f(x)\big)^{2024} \ \text{ for } x \in \mathbb{R} \right\}.$$

Then, the number of elements in S is ______________

Q.47 Let $a_1 = 1, b_1 = 2$ and $c_1 = 3$. Consider the convergent sequences

$$\{a_n\}_{n=1}^{\infty}\ ,\ \{b_n\}_{n=1}^{\infty} \text{ and } \{c_n\}_{n=1}^{\infty}$$

defined as follows:

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \frac{b_n + c_n}{2} \text{ and } c_{n+1} = \frac{c_n + a_n}{2} \text{ for } n \geq 1.$$

Then,

$$\sum_{n=1}^{\infty} b_n c_n (a_{n+1} - a_n) + \sum_{n=1}^{\infty} (b_{n+1}c_{n+1} - b_n c_n) a_{n+1}$$

equals __________ (*rounded off to two decimal places*)

Q.48 Let

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 4, (x-1)^2 + y^2 \leq 1, z \geq 0\}.$$

Then, the surface area of S equals ___________ (*rounded off to two decimal places*).

Q.49 Let $P_7(x)$ be the real vector space of polynomials in x with degree at most 7, together with the zero polynomial. For $r = 1, 2, \dots, 7,$ define

$$s_r(x) = x(x-1)\cdots\big(x-(r-1)\big) \text{ and } s_0(x) = 1.$$

Consider the fact that $B = \{s_0(x), s_1(x), \dots, s_7(x)\}$ is a basis of $P_7(x)$.

If

$$x^5 = \sum_{k=0}^{7} \alpha_{5,k}\, s_k(x)\,,$$

where $\alpha_{5,k} \in \mathbb{R}$, then $\alpha_{5,2}$ equals ___________ (*rounded off to two decimal places*)

Q.50 Let

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}.$$

If $p(x)$ is the characteristic polynomial of M, then $p(2) - 1$ equals __________

Section C: Q.51 – Q.60 Carry TWO marks each.

Q.51 For $\alpha \in (-2\pi, 0)$, consider the differential equation

$$x^2 \frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + y = 0 \quad \text{for} \quad x > 0.$$

Let D be the set of all $\alpha \in (-2\pi, 0)$ for which all corresponding real solutions to the above differential equation approach zero as $x \to 0^+$. Then, the number of elements in $D \cap \mathbb{Z}$ equals ___________

Q.52 The value of

$$\lim_{t\to\infty}\left(\left(\log\left(t^2 + \frac{1}{t^2}\right)\right)^{-1} \int_1^{\pi t} \frac{\sin^2 5x}{x}\,dx\right)$$

equals ____________ (*rounded off to two decimal places*).

Q.53 Let T be the planar region enclosed by the square with vertices at the points $(0,1), (1,0), (0,-1)$ and $(-1,0)$. Then, the value of

$$\iint_T \Big(\cos(\pi(x-y)) - \cos(\pi(x+y))\Big)^2 \, dx\, dy$$

equals ____________ (*rounded off to two decimal places*).

Q.54 Let

$$S = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 < 1\}.$$

Then, the value of

$$\frac{1}{\pi} \iiint_S \left((x-2y+z)^2 + (2x-y-z)^2 + (x-y+2z)^2 \right) dxdydz$$

equals ________ (*rounded off to two decimal places*).

Q.55 For $n \in \mathbb{N}$, if

$$a_n = \frac{1}{n^3+1} + \frac{2^2}{n^3+2} + \cdots + \frac{n^2}{n^3+n}$$

then the sequence $\{a_n\}_{n=1}^{\infty}$ converges to ____________ (*rounded off to two decimal places*)

Q.56 Consider the function $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3 - 4x^2 + 4x - 6$.

For $c \in \mathbb{R}$, let

$$S(c) = \left\{ x \in \mathbb{R} : f(x) = c \right\}$$

and $|S(c)|$ denote the number of elements in $S(c)$. Then, the value of

$$|S(-7)| + |S(-5)| + |S(3)|$$

equals _________

Q.57 Let $c > 0$ be such that

$$\int_0^c e^{s^2}\, ds = 3$$

Then, the value of

$$\int_0^c \left(\int_x^c e^{x^2+y^2}\, dy \right) dx$$

equals ___________(*rounded off to one decimal place*).

Q.58 For $k \in \mathbb{N}$, let $0 = t_0 < t_1 < \cdots < t_k < t_{k+1} = 1$. A function $f: [0,1] \to \mathbb{R}$ is said to be piecewise linear with nodes $t_1, \ldots, t_k$, if for each $j = 1, 2, \ldots, k+1$, there exist $a_j \in \mathbb{R}$, $b_j \in \mathbb{R}$ such that

$$f(t) = a_j + b_j t \text{ for } t_{j-1} < t < t_j .$$

Let V be the real vector space of all real valued continuous piecewise linear functions on $[0,1]$ with nodes $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. Then, the dimension of V equals ______________

Q.59 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{n^{n-1}} \sum_{k=0}^{n} \frac{n!}{k!\,(n-k)!} \frac{n^k}{k+1}$$

and $\beta = \lim_{n\to\infty} a_n$. Then, the value of $\log \beta$ equals ___________ (*rounded off to two decimal places*).

Q.60 Define the function $f: (-1,1) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by

$$f(x) = \sin^{-1} x$$

Let a_6 denote the coefficient of x^6 in the Taylor series of $\left(f(x)\right)^2$ about $x = 0$. Then, the value of $9a_6$ equals ____________ (*rounded off to two decimal places*).

JAM 2024: Mathematics (MA)
Master Answer Key

Q. No.	Session	Question Type	Section	Key/Range*	Marks
1	1	MCQ	A	B	1
2	1	MCQ	A	A	1
3	1	MCQ	A	A	1
4	1	MCQ	A	A	1
5	1	MCQ	A	A	1
6	1	MCQ	A	D	1
7	1	MCQ	A	A	1
8	1	MCQ	A	A	1
9	1	MCQ	A	D	1
10	1	MCQ	A	B	1
11	1	MCQ	A	A	2
12	1	MCQ	A	B	2
13	1	MCQ	A	A	2
14	1	MCQ	A	D	2
15	1	MCQ	A	A	2
16	1	MCQ	A	A	2
17	1	MCQ	A	B	2
18	1	MCQ	A	A	2
19	1	MCQ	A	D	2
20	1	MCQ	A	A	2
21	1	MCQ	A	A	2
22	1	MCQ	A	D	2
23	1	MCQ	A	A	2
24	1	MCQ	A	A	2
25	1	MCQ	A	D	2
26	1	MCQ	A	B	2
27	1	MCQ	A	A	2
28	1	MCQ	A	B	2
29	1	MCQ	A	C	2
30	1	MCQ	A	A	2
31	1	MSQ	B	A;B	2
32	1	MSQ	B	A;B	2
33	1	MSQ	B	A	2
34	1	MSQ	B	B;D	2

JAM 2024: Mathematics (MA)
Master Answer Key

Q. No.	Session	Question Type	Section	Key/Range*	Marks
35	1	MSQ	B	B;C	2
36	1	MSQ	B	B;C	2
37	1	MSQ	B	B;C	2
38	1	MSQ	B	B;D	2
39	1	MSQ	B	B;C	2
40	1	MSQ	B	B;D	2
41	1	NAT	C	0.24 to 0.26	1
42	1	NAT	C	0.19 to 0.21	1
43	1	NAT	C	1.0 to 1.0	1
44	1	NAT	C	0.15 to 0.18	1
45	1	NAT	C	6 to 6	1
46	1	NAT	C	3 to 3	1
47	1	NAT	C	1.95 to 2.05	1
48	1	NAT	C	4.50 to 4.60	1
49	1	NAT	C	14.95 to 15.05	1
50	1	NAT	C	31 to 31	1
51	1	NAT	C	6 to 6	2
52	1	NAT	C	0.24 to 0.26	2
53	1	NAT	C	1.95 to 2.05	2
54	1	NAT	C	4.70 to 4.90	2
55	1	NAT	C	0.30 to 0.40	2
56	1	NAT	C	5 to 5	2
57	1	NAT	C	4.4 to 4.6	2
58	1	NAT	C	5 to 5	2
59	1	NAT	C	0.98 to 1.02	2
60	1	NAT	C	1.50 to 1.70	2

Paper Specific Instructions

1. The examination is of 3 hours duration. There are a total of 60 questions carrying 100 marks. The entire paper is divided into three sections, **A**, **B** and **C**. All sections are compulsory. Questions in each section are of different types.

2. **Section A** contains a total of 30 **Multiple Choice Questions (MCQ).** Each MCQ type question has four choices out of which only **one** choice is the correct answer. Questions Q.1 – Q.30 belong to this section and carry a total of 50 marks. Q.1 – Q.10 carry 1 mark each and Questions Q.11 – Q.30 carry 2 marks each.

3. **Section B** contains a total of 10 **Multiple Select Questions (MSQ).** Each MSQ type question is similar to MCQ but with a difference that there will be **more than one** choices that are correct out of the four given choices. The candidate gets full credit if he/she selects all the correct answers only and no wrong answers. Questions Q.31 – Q.40 belong to this section and carry 2 marks each with a total of 20 marks.

4. **Section C** contains a total of 20 **Numerical Answer Type (NAT)** questions. For these NAT type questions, the answer is a real number which needs to be entered using the virtual keyboard on the monitor. No choices will be shown for these type of questions. Questions Q.41 – Q.60 belong to this section and carry a total of 30 marks. Q.41 – Q.50 carry 1 mark each and Questions Q.51 – Q.60 carry 2 marks each.

5. In all sections, questions not attempted will result in zero mark. In **Section A** (MCQ), wrong answer will result in **NEGATIVE** marks. For all 1-mark questions, 1/3 marks will be deducted for each wrong answer. For all 2-mark questions, 2/3 marks will be deducted for each wrong answer. In **Section B** (MSQ), there is **NO NEGATIVE** and **NO PARTIAL** marking provisions. There is **NO NEGATIVE** marking in **Section C** (NAT) as well.

6. Only Virtual Scientific Calculator is allowed. Charts, graph sheets, tables, cellular phone or other electronic gadgets are **NOT** allowed in the examination hall.

7. A Scribble Pad will be provided for rough work.

Special Instructions / Useful Data

$\mathbb{N}$ = The set of all natural numbers

$\mathbb{Z}$ = The set of all integers

$\mathbb{Z}_n = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$, the group of integers modulo n, under addition modulo n, for $n \in \mathbb{N}$

$\mathbb{R}$ = The set of all real numbers

$\mathbb{R}^n =$ The n −dimensional Euclidean space

ln x = The natural logarithm of x (to the base e)

S_n = The symmetric group of all permutations on $\{1,2, \ldots, n\}$

id = The identity element in S_n

(a_n) = The infinite sequence $a_1, a_2, a_3, \ldots$

$f \circ g$ = The composition of f and g, defined by $(f \circ g)(x) = f(g(x))$

$f'(x)$ = The first derivative of f at the point x

$f''(x)$ = The second derivative of f at the point x

span S = The linear span of the subset S of a vector space

$P_n(\mathbb{R})$ = The real vector space of real polynomials of degree less than or equal to n, together with the zero polynomial. These polynomials can be regarded as functions from $\mathbb{R}$ to $\mathbb{R}$

$\ker(T)$ = The kernel of the linear transformation T

$M = (m_{ij})$ = Matrix of appropriate order with the entry/element in the i^{th} row and j^{th} column denoted by m_{ij}, $m_{ij} \in \mathbb{R}$

$\gcd(m, n)$ = The greatest common divisor of the natural numbers m and n

$\det(M)$ = The determinant of the matrix M

$\frac{\partial f}{\partial x}$ = The partial derivative of f with respect to x

$\frac{\partial f}{\partial y}$ = The partial derivative of f with respect to y

Section A: Q.1 – Q.10 Carry ONE mark each.

Q.1 The sum of the infinite series

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\pi^{2n+1}}{2^{2n+1}(2n)!}$$

is equal to

(A) $-\pi$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $-\frac{\pi}{4}$

Q.2 For which one of the following choices of $N(x, y)$, is the equation

$$(e^x \sin y - 2y \sin x)\, \mathrm{d}x + N(x, y)\, \mathrm{d}y = 0$$

an exact differential equation?

(A) $N(x, y) = e^x \sin y + 2 \cos x$

(B) $N(x, \mathrm{y}) = e^x \cos y + 2 \cos x$

(C) $N(x, y) = e^x \cos y + 2 \sin x$

(D) $N(x, y) = e^x \sin y + 2 \sin x$

Q.3 Let $f\ ,\ g : \mathbb{R} \to \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} x\,|x|\left|\sin\dfrac{1}{x}\right| & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} x^2 \sin\dfrac{1}{x} + x\cos\dfrac{1}{x} & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0\,. \end{cases}$$

Then, which one of the following is TRUE?

(A) f is differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

(B) f is NOT differentiable at $x = 0$, and g is differentiable at $x = 0$

(C) f is differentiable at $x = 0$, and g is differentiable at $x = 0$

(D) f is NOT differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

Q.4 Let $f,\ g : \mathbb{R} \to \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} |x|^{1/8} \left|\sin \frac{1}{x}\right| \cos x & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} e^x \cos \frac{1}{x} & \text{if} \quad x \neq 0 \\ 1 & \text{if} \quad x = 0 \end{cases}.$$

Then, which one of the following is TRUE?

(A) f is continuous at $x = 0$, and g is NOT continuous at $x = 0$

(B) f is NOT continuous at $x = 0$, and g is continuous at $x = 0$

(C) f is continuous at $x = 0$, and g is continuous at $x = 0$

(D) f is NOT continuous at $x = 0$, and g is NOT continuous at $x = 0$

Q.5 Which one of the following is the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x} \ ?$$

(A) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(B) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(C) $\alpha_1 e^{-4x} + \alpha_2 e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(D) $\alpha_1 x e^{-4x} + \alpha_2 x^2 e^{-4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

Q.6 Define $T: \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T(x, y, z) = (x + z,\ 2x + 3y + 5z,\ 2y + 2z), \text{ for all } (x, y, z) \in \mathbb{R}^3 .$$

Then, which one of the following is TRUE?

(A) T is one-one and T is NOT onto

(B) T is NOT one-one and T is onto

(C) T is one-one and T is onto

(D) T is NOT one-one and T is NOT onto

Q.7 Let $M = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & x \end{pmatrix}$ for some real number x.

If 0 is an eigenvalue of M, then $(M^4 + M)\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is equal to

(A) $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$

(D) $\begin{pmatrix} 17 \\ 0 \\ 17 \end{pmatrix}$

Q.8 Let $\mathrm{T} : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ be the linear transformation defined by $\mathrm{T}\big(p(x)\big) = p(x+1)$, for all $p(x) \in P_2(\mathbb{R})$. If M is the matrix representation of T with respect to the ordered basis $\{1, x, x^2\}$ of $P_2(\mathbb{R})$, then which one of the following is TRUE?

(A) The determinant of M is 2

(B) The rank of M is 2

(C) 1 is the only eigenvalue of M

(D) The nullity of M is 2

Q.9 Let G be a finite abelian group of order 10. Let x_0 be an element of order 2 in G.

If $X = \{x \in G : x^3 = x_0\}$, then which one of the following is TRUE?

(A) X has exactly one element

(B) X has exactly two elements

(C) X has exactly three elements

(D) X is an empty set

Q.10 The value of

$$\int_0^1 \left(\int_{\sqrt{y}}^1 3e^{x^3}\, dx \right) dy$$

is equal to

(A) $e - 1$

(B) $\frac{e-1}{2}$

(C) $\sqrt{e} - 1$

(D) $\frac{\sqrt{e}-1}{2}$

Section A: Q.11 – Q.30 Carry TWO marks each.

Q.11 Let $\mathcal{C}$ denote the family of curves described by $yx^2 = \lambda$, for $\lambda \in (0, \infty)$ and lying in the first quadrant of the xy plane. Let $\mathcal{O}$ denote the family of orthogonal trajectories of $\mathcal{C}$.

Which one of the following curves is a member of $\mathcal{O}$, and passes through the point $(2, 1)$?

(A) $y = \frac{x^2}{4}$, $x > 0$, $y > 0$

(B) $x^2 - 2y^2 = 2$, $x > 0$, $y > 0$

(C) $x - y = 1$, $x > 0$, $y > 0$

(D) $2x - y^2 = 3$, $x > 0$, $y > 0$

Q.12 Let $\varphi : (0, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$x\frac{dy}{dx} = (\ln y - \ln x)y,$$

satisfying $\varphi(1) = e^2$. Then, the value of $\varphi(2)$ is equal to

(A) e^2

(B) $2e^3$

(C) $3e^2$

(D) $6e^3$

Q.13 Let $X = \{x \in S_4 : x^3 = id\}$ and $Y = \{x \in S_4 : x^2 \neq id\}$.

If m and n denote the number of elements in X and Y, respectively, then which one of the following is TRUE?

(A) m is even and n is even

(B) m is odd and n is even

(C) m is even and n is odd

(D) m is odd and n is odd

Q.14 Let $\varphi : \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} = (y-1)\,(y-3)\,,$$

satisfying $\varphi(0) = 2$. Then, which one of the following is TRUE?

(A) $\lim_{x\to\infty} \varphi(x) = 0$

(B) $\lim_{x\to \ln\sqrt{2}} \varphi(x) = 1$

(C) $\lim_{x\to -\infty} \varphi(x) = 3$

(D) $\lim_{x\to \ln\frac{1}{\sqrt{2}}} \varphi(x) = 6$

Q.15 Let $M = \begin{pmatrix} 6 & 2 & -6 & 8 \\ 5 & 3 & -9 & 8 \\ 3 & 1 & -2 & 4 \end{pmatrix}$. Consider the system S of linear equations given by

$$\begin{aligned} 6x_1 + 2x_2 - 6x_3 + 8x_4 &= 8 \\ 5x_1 + 3x_2 - 9x_3 + 8x_4 &= 16 \\ 3x_1 + x_2 - 2x_3 + 4x_4 &= 32 \end{aligned}$$

where x_1, x_2, x_3, x_4 are unknowns.

Then, which one of the following is TRUE?

(A) The rank of M is 3, and the system S has a solution

(B) The rank of M is 3, and the system S does NOT have a solution

(C) The rank of M is 2, and the system S has a solution

(D) The rank of M is 2, and the system S does NOT have a solution

Q.16 Let $M = \begin{pmatrix} -2 & 0 & 0 \\ 3 & 2 & 3 \\ 4 & -1 & x \end{pmatrix}$ for some real number x. Suppose that -2 and 3 are eigenvalues of M. If $M^3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 125 \\ 125 \end{pmatrix}$, then which one of the following is TRUE?

(A) $x = 5$, and the matrix $M^2 + M$ is invertible

(B) $x \neq 5$, and the matrix $M^2 + M$ is invertible

(C) $x = 5$, and the matrix $M^2 + M$ is NOT invertible

(D) $x \neq 5$, and the matrix $M^2 + M$ is NOT invertible

Q.17 Let $f(x) = 10x^2 + e^x - \sin(2x) - \cos x,\ x \in \mathbb{R}$. The number of points at which the function f has a local minimum is

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Q.18 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \cos\frac{1}{n} \quad \text{and} \quad y_n = \sum_{k=1}^{n} \frac{1}{n+k}.$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

(B) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ converges

(C) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ converges

(D) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

Q.19 Let $x_1 = \frac{5}{2}$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \frac{1}{5}(x_n^2 + 6).$$

Then, which one of the following is TRUE?

(A) (x_n) is an increasing sequence, and (x_n) is NOT a bounded sequence

(B) (x_n) is NOT an increasing sequence, and (x_n) is NOT a bounded sequence

(C) (x_n) is NOT a decreasing sequence, and (x_n) is a bounded sequence

(D) (x_n) is a decreasing sequence, and (x_n) is a bounded sequence

Q.20 Let $x_1 = 2$ and $x_{n+1} = 2 + \frac{1}{2x_n}$ for all $n \in \mathbb{N}$.

Then, which one of the following is TRUE?

(A) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence

(B) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence

(C) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

(D) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

Q.21 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \frac{3^n}{n^3} \text{ and } y_n = \left(4^n + (-1)^n 3^n\right)^{1/n}.$$

Then, which one of the following is TRUE?

(A) (x_n) has a convergent subsequence, and NO subsequence of (y_n) is convergent

(B) NO subsequence of (x_n) is convergent, and (y_n) has a convergent subsequence

(C) (x_n) has a convergent subsequence, and (y_n) has a convergent subsequence

(D) NO subsequence of (x_n) is convergent, and NO subsequence of (y_n) is convergent

Q.22 Let $M = (m_{ij})$ be a 3×3 real, invertible matrix and $\sigma \in S_3$ be the permutation defined by $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$. The matrix $M_\sigma = (n_{ij})$ is defined by $n_{ij} = m_{i\sigma(j)}$ for all $i, j \in \{1, 2, 3\}$.

Then, which one of the following is TRUE?

(A) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

(B) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1

(C) $\det(M) = \det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 1

(D) $\det(M) = -\det(M_\sigma)$, and nullity of the matrix $M - M_\sigma$ is 0

Q.23 Let $\mathbb{R}/\mathbb{Z}$ denote the quotient group, where $\mathbb{Z}$ is considered as a subgroup of the additive group of real numbers $\mathbb{R}$.

Let m denote the number of injective (one-one) group homomorphisms from $\mathbb{Z}_3$ to $\mathbb{R}/\mathbb{Z}$ and n denote the number of group homomorphisms from $\mathbb{R}/\mathbb{Z}$ to $\mathbb{Z}_3$.

Then, which one of the following is TRUE?

(A) $m = 2$ and $n = 1$

(B) $m = 3$ and $n = 3$

(C) $m = 2$ and $n = 3$

(D) $m = 1$ and $n = 1$

Q.24 Let f_1, f_2, f_3 be nonzero linear transformations from $\mathbb{R}^4$ to $\mathbb{R}$ and

$$\ker(f_1) \subseteq \ker(f_2) \cap \ker(f_3) .$$

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by

$$T(v) = \big(f_1(v), f_2(v), f_3(v)\big), \qquad \text{for all } v \in \mathbb{R}^4 .$$

Then, the nullity of T is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Q.25 Let $x_1 = 1$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \left(\frac{1}{2} + \frac{\sin^2 n}{n}\right) x_n .$$

Then, which one of the following is TRUE?

(A) $\sum_{n=1}^{\infty} x_n$ converges

(B) $\sum_{n=1}^{\infty} x_n$ does NOT converge

(C) $\sum_{n=1}^{\infty} x_n^2$ does NOT converge

(D) $\sum_{n=1}^{\infty} x_n x_{n+1}$ does NOT converge

Q.26 Let $x_1 > 0$. For $n \in \mathbb{N}$, define $x_{n+1} = x_n + 4$. If

$$\lim_{n\to\infty}\left(\frac{1}{x_2x_3} + \frac{1}{x_3x_4} + \cdots + \frac{1}{x_{n+1}x_{n+2}}\right) = \frac{1}{24},$$

then the value of x_1 is equal to

(A) 1

(B) 2

(C) 3

(D) 8

Q.27 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = e^y(x^2 + y^2) \quad \text{for all } (x,y) \in \mathbb{R}^2 .$$

Then, which one of the following is TRUE?

(A) The number of points at which f has a local minimum is 2

(B) The number of points at which f has a local maximum is 2

(C) The number of points at which f has a local minimum is 1

(D) The number of points at which f has a local maximum is 1

Q.28 Let Ω be the bounded region in $\mathbb{R}^3$ lying in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$), and bounded by the surfaces $z = x^2 + y^2$, $z = 4$, $x = 0$ and $y = 0$.

Then, the volume of Ω is equal to

(A) π

(B) 2π

(C) 3π

(D) 4π

Q.29 Let $\varphi : [0,\infty) \to \mathbb{R}$ be the continuous function satisfying

$$\varphi(x) = \left(\int_0^x \varphi(t)dt\right) + \sin x \ , \ \text{ for all } x \in [0,\infty)\,.$$

Then, the value of $\lim\limits_{x\to\pi/2} \left(2\varphi(x) - e^x\right)$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Q.30 The number of elements in the set

$$\{x \in \mathbb{R} : 8x^2 + x^4 + x^8 = \cos x\}$$

is equal to

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Section B: Q.31 – Q.40 Carry TWO marks each.

Q.31 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{xy^2 + y^5}{x^2 + y^4} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Then, which of the following is/are TRUE?

(A) The iterated limits $\lim_{x \to 0} \left(\lim_{y \to 0} f(x,y) \right)$ and $\lim_{y \to 0} \left(\lim_{x \to 0} f(x,y) \right)$ exist

(B) Exactly one of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists at $(0,0)$

(C) Both the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at $(0,0)$

(D) f is NOT differentiable at $(0,0)$

Q.32 If $M, N, \mu, w: \mathbb{R}^2 \to \mathbb{R}$ are differentiable functions with continuous partial derivatives, satisfying

$$\mu(x,y)M(x,y)dx + \mu(x,y)N(x,y)dy = dw$$

then which of the following is/are TRUE?

(A) μw is an integrating factor for $M(x,y)dx + N(x,y)dy = 0$

(B) μw^2 is an integrating factor for $M(x,y)dx + N(x,y)dy = 0$

(C) $w(x,y) = w(0,0) + \int_0^x (\mu M)(s,0)ds + \int_0^y (\mu N)(x,t)dt$, for all $(x,y) \in \mathbb{R}^2$

(D) $w(x,y) = w(0,0) + \int_0^x (\mu M)(s,y)ds + \int_0^x (\mu N)(0,t)dt$, for all $(x,y) \in \mathbb{R}^2$

Q.33 Let $\varphi : (-1, \infty) \to (0, \infty)$ be the solution of the differential equation

$$\frac{dy}{dx} - 2\,ye^{x} = 2e^{x}\sqrt{y}\,,$$

satisfying $\varphi(0) = 1$.

Then, which of the following is/are TRUE?

(A) φ is an unbounded function

(B) $\lim_{x \to \ln 2} \varphi(x) = (2e - 1)^2$

(C) $\lim_{x \to \ln 2} \varphi(x) = \sqrt{2e - 1}$

(D) φ is a strictly increasing function on the interval $(0, \infty)$

Q.34 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{(x^2 + \sin x)y^2}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Then, which of the following is/are TRUE?

(A) $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exists and $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 1$

(B) $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exists and $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$

(C) f is differentiable at $(0,0)$

(D) f is NOT differentiable at $(0,0)$

Q.35 Let

$$u_1 = (1,0,0,-1), u_2 = (0,2,0,-1),\ \ u_3 = (0,0,1,-1) \text{ and } u_4 = (0,0,0,1)$$

be elements in the real vector space $\mathbb{R}^4$.

Then, which of the following is/are TRUE?

(A) $\{u_1, u_2, u_3, u_4\}$ is a linearly independent set in $\mathbb{R}^4$

(B) $\{u_1 - u_2, u_2 - u_3, u_3 - u_4, u_4 - u_1\}$ is NOT a linearly independent set in $\mathbb{R}^4$

(C) $\{u_1, -u_2, u_3, -u_4\}$ is NOT a linearly independent set in $\mathbb{R}^4$

(D) $\{u_1 + u_2, u_2 + u_3, u_3 + u_4, u_4 + u_1\}$ is a linearly independent set in $\mathbb{R}^4$

Q.36 For $n \in \mathbb{N}$, let

$$x_n = \sum_{k=1}^{n} \frac{k}{n^2 + k}.$$

Then, which of the following is/are TRUE?

(A) The sequence (x_n) converges

(B) The series $\sum_{n=1}^{\infty} x_n$ converges

(C) The series $\sum_{n=1}^{\infty} x_n$ does NOT converge

(D) The series $\sum_{n=1}^{\infty} x_n^n$ converges

Q.37 Let $f: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 0, f'(0) = 2 \text{ and } f(1) = -3\,.$$

Then, which of the following is/are TRUE?

(A) $|f'(x)| \leq 2$ for all $x \in [0,1]$

(B) $|f'(x_1)| > 2$ for some $x_1 \in [0,1]$

(C) $|f''(x)| < 10$ for all $x \in [0,1]$

(D) $|f''(x_2)| \geq 10$ for some $x_2 \in [0,1]$

Q.38 Let $f: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 4, f(1) = -2, f(2) = 8 \text{ and } f(3) = 2\,.$$

Then, which of the following is/are TRUE?

(A) $|f'(x)| < 5$ for all $x \in [0, 1]$

(B) $|f'(x_1)| \geq 5$ for some $x_1 \in [0, 1]$

(C) $f'(x_2) = 0$ for some $x_2 \in [0, 3]$

(D) $f''(x_3) = 0$ for some $x_3 \in [0, 3]$

Q.39 For $n \in \mathbb{N}$, consider the set $U(n) = \{\bar{x} \in \mathbb{Z}_n : \gcd(x, n) = 1\}$ as a group under multiplication modulo n.

Then, which of the following is/are TRUE?

(A) $U(8)$ is a cyclic group

(B) $U(5)$ is a cyclic group

(C) $U(12)$ is a cyclic group

(D) $U(9)$ is a cyclic group

Q.40 Consider the following subspaces of the real vector space $\mathbb{R}^3$:

$V_1 = \text{span}\,\{(1,2,3),(1,1,0)\}$,

$V_2 = \text{span}\,\{(1,-1,0)\}$,

$V_3 = \text{span}\,\{(1,1,1)\}$,

$V_4 = \text{span}\,\{(1,3,6)\}$ and

$V_5 = \text{span}\,\{(1,0,-3)\}$.

Then, which of the following is/are TRUE?

(A) $V_1 \cup V_2$ is a subspace of $\mathbb{R}^3$

(B) $V_1 \cup V_3$ is a subspace of $\mathbb{R}^3$

(C) $V_1 \cup V_4$ is a subspace of $\mathbb{R}^3$

(D) $V_1 \cup V_5$ is a subspace of $\mathbb{R}^3$

Section C: Q.41 – Q.50 Carry ONE mark each.

Q.41 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{\left(x+\frac{1}{4}\right)^n}{(-2)^n n^2}$$

about $x = -\frac{1}{4}$, is equal to ______________ *(rounded off to two decimal places).*

Q.42 The value of

$$\lim_{n\to\infty} 8n \left(e^{\left(\frac{1}{2n}\right)} - 1\right)\left(\sin\frac{1}{2n} + \left|\cos\frac{1}{2n}\right|\right)$$

is equal to ________________ *(rounded off to two decimal places).*

Q.43 Let α be the real number such that

$$\lim_{x\to 0} \frac{(1-\cos x)(2^{2+x}-4)}{x^3} = \alpha \ln 2 \ .$$

Then, the value of α is equal to ____________ *(rounded off to two decimal places).*

Q.44 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$4\frac{d^2y}{dx^2} + 16\,\frac{dy}{dx} + 25y = 0$$

satisfying $\varphi(0) = 1$ and $\varphi'(0) = -\frac{1}{2}$.

Then, the value of $\lim_{x\to\pi/6} e^{2x}\varphi(x)$ is equal to ______________ *(rounded off to two decimal places).*

Q.45 Let S be the surface area of the portion of the plane $z = x + y + 3$, which lies inside the cylinder $x^2 + y^2 = 1$.

Then, the value of $\left(\frac{S}{\pi}\right)^2$ is equal to _____________ *(rounded off to two decimal places).*

Q.46 Consider the following subspaces of $\mathbb{R}^4$:

$$V_1 = \{(x, y, z, w) \in \mathbb{R}^4 : x + y + 2w = 0\},$$

$$V_2 = \{(x, y, z, w) \in \mathbb{R}^4 : 2y + z + w = 0\},$$

$$V_3 = \{(x, y, z, w) \in \mathbb{R}^4 : x + 3y + z + 3w = 0\}.$$

Then, the dimension of the subspace $V_1 \cap V_2 \cap V_3$ is equal to ___________.

Q.47 Consider the real vector space $\mathbb{R}^3$. Let $T : \mathbb{R}^3 \to \mathbb{R}$ be a linear transformation such that

$$T(1,1,1) = 0, \qquad T(1,-1,1) = 0 \text{ and } T(0,0,1) = 16.$$

Then, the value of $T\left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right)$ is equal to ____________________ *(rounded off to two decimal places).*

Q.48 Let T denote the triangle in the xy plane bounded by the x axis and the lines $y = x$ and $x = 1$. The value of the double integral (over T)

$$\iint_T (5 - y)dxdy$$

is equal to __________ *(rounded off to two decimal places).*

Q.49 Let $T, S : P_4(\mathbb{R}) \to P_4(\mathbb{R})$ be the linear transformations defined by

$$T\big(p(x)\big) = xp'(x) \text{ and } S\big(p(x)\big) = (x+1)p'(x)$$

for all $p(x) \in P_4(\mathbb{R})$.

Then, the nullity of the composition $S \circ T$ is _____________.

Q.50 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{(x^2 - y^2)xy}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Then, the value of $\frac{\partial f}{\partial y}(1,0) - \frac{\partial f}{\partial x}(0,2)$ is equal to ____________ *(rounded off to two decimal places).*

Section C: Q.51 – Q.60 Carry TWO marks each.

Q.51 Let $f: \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying

$$\int_0^{\pi/4} \left(\sin(x)\, f(x) \; + \cos(x) \int_0^x f(t)dt \right) dx = \sqrt{2}\,.$$

Then, the value of $\int_0^{\pi/4} f(x)\, dx$ is equal to ________________ *(rounded off to two decimal places).*

Q.52 Let $\sigma \in S_4$ be the permutation defined by $\sigma(1) = 2,\ \sigma(2) = 3,\ \sigma(3) = 1$ and $\sigma(4) = 4$. The number of elements in the set

$$\{\tau \in S_4 : \tau\sigma\tau^{-1} = \sigma\}$$

is equal to _____________.

Q.53 Let $f(x) = 2x - \sin x$, for all $x \in \mathbb{R}$. Let $k \in \mathbb{N}$ be such that

$$\lim_{x \to 0} \left(\frac{1}{x} \sum_{i=1}^{k} i^2 f\left(\frac{x}{i}\right) \right) = 45.$$

Then, the value of k is equal to ______________.

Q.54 The value of the infinite series

$$\sum_{n=1}^{\infty} n \left(\frac{3}{4}\right)^{2(n-1)}$$

is equal to _______________ *(rounded off to two decimal places)*

Q.55 Let $\varphi: (0, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 6x \ln x$$

satisfying $\varphi(1) = -3$ and $\varphi(e) = 0$.

Then, the value of $|\varphi'(1)|$ is equal to ______________ *(rounded off to two decimal places).*

Q.56 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} + 2xy = 2 + 4x^2$$

satisfying $\varphi(0) = 0$.

Then, the value of $\varphi(2)$ is equal to ________________ *(rounded off to two decimal places).*

Q.57 Let Ω be the solid bounded by the planes $z = 0$, $y = 0$, $x = \frac{1}{2}$, $2y = x$ and $2x + y + z = 4$.

If V is the volume of Ω, then the value of $64\,V$ is equal to ____________ *(rounded off to two decimal places).*

Q.58 Let the subspace H of $P_3(\mathbb{R})$ be defined as

$$H = \{p(x) \in P_3(\mathbb{R}) : xp'(x) = 3p(x)\}.$$

Then, the dimension of H is equal to ________________.

Q.59 Let G be an abelian group of order 35. Let m denote the number of elements of order 5 in G, and let n denote the number of elements of order 7 in G.

Then, the value of $m + n$ is equal to ______________.

Q.60 The number of surjective (onto) group homomorphisms from S_4 to $\mathbb{Z}_6$ is equal to ________________.

Q. No.	Question Type	Subject Name	Key/Range	Mark(s)
1	MCQ	MA	C	1
2	MCQ	MA	B	1
3	MCQ	MA	A	1
4	MCQ	MA	A	1
5	MCQ	MA	A	1
6	MCQ	MA	D	1
7	MCQ	MA	B	1
8	MCQ	MA	C	1
9	MCQ	MA	A	1
10	MCQ	MA	A	1
11	MCQ	MA	B	2
12	MCQ	MA	B	2
13	MCQ	MA	B	2
14	MCQ	MA	C	2
15	MCQ	MA	A	2
16	MCQ	MA	B	2
17	MCQ	MA	B	2
18	MCQ	MA	D	2
19	MCQ	MA	D	2
20	MCQ	MA	A	2
21	MCQ	MA	B	2

Q. No.	Question Type	Subject Name	Key/Range	Mark(s)
22	MCQ	MA	C	2
23	MCQ	MA	A	2
24	MCQ	MA	C	2
25	MCQ	MA	A	2
26	MCQ	MA	B	2
27	MCQ	MA	C	2
28	MCQ	MA	B	2
29	MCQ	MA	A	2
30	MCQ	MA	C	2
31	MSQ	MA	A;C;D	2
32	MSQ	MA	A;B;C	2
33	MSQ	MA	A;B;D	2
34	MSQ	MA	B;D	2
35	MSQ	MA	A;B	2
36	MSQ	MA	A;C;D	2
37	MSQ	MA	B;D	2
38	MSQ	MA	B;C;D	2
39	MSQ	MA	B;D	2
40	MSQ	MA	C;D	2
41	NAT	MA	1.98 to 2.02	1
42	NAT	MA	3.98 to 4.02	1

Q. No.	Question Type	Subject Name	Key/Range	Mark(s)
43	NAT	MA	1.98 to 2.02	1
44	NAT	MA	1.39 to 1.43	1
45	NAT	MA	2.98 to 3.02	1
46	NAT	MA	2 to 2	1
47	NAT	MA	3.98 to 4.02	1
48	NAT	MA	2.31 to 2.35	1
49	NAT	MA	1 to 1	1
50	NAT	MA	2.98 to 3.02	1
51	NAT	MA	1.98 to 2.02	2
52	NAT	MA	3 to 3	2
53	NAT	MA	9 to 9	2
54	NAT	MA	5.20 to 5.24	2
55	NAT	MA	0.98 to 1.02	2
56	NAT	MA	3.98 to 4.02	2
57	NAT	MA	MTA	2
58	NAT	MA	1 to 1	2
59	NAT	MA	10 to 10	2
60	NAT	MA	0 to 0	2

Special Instructions / Useful Data

$\mathbb{Z}$ = The set of all integers
$\mathbb{Q}$ = The set of all rational numbers
$\mathbb{R}$ = The set of all real numbers
$\mathbb{C}$ = The set of all complex numbers
$\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$, where $\mathbb{R}$ occurs n times
$\mathbb{C}^n = \mathbb{C} \times \mathbb{C} \times \cdots \times \mathbb{C}$, where $\mathbb{C}$ occurs n times
S_n = The symmetric group of all permutations on $1,2,3, \dots, n$
$\mathbb{Z}_n$ = The additive group of integers modulo n
$M_{m,n}(\mathbb{R})$ = The set of all $m \times n$ matrices with real entries
$M_{m,n}(\mathbb{C})$ = The set of all $m \times n$ matrices with complex entries
$M_n(\mathbb{R})$ = The set of all $n \times n$ matrices with real entries
$M_n(\mathbb{C})$ = The set of all $n \times n$ matrices with complex entries
M^T = The transpose of the matrix M
M^* = The conjugate transpose of the matrix M
$A \setminus B$ = The set of all elements of A which are not in B, for sets A and B
$[x]$ = The largest integer less than or equal to x for any real number x
$f'(x) = \frac{df}{dx}$
$f^{(k)}(x) = \frac{d^k f}{dx^k}$
$g \circ f$ is the composition function given by $(g \circ f)(x) = g(f(x))$

Section A: Q.1 – Q.10 Carry ONE mark each.	
Q.1	For each positive integer n, let $x_n = 1 - (-1)^n + \frac{1}{n}$ and $y_n = \left(1 + \frac{1}{2n}\right)^{3n}$. Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is not convergent and (y_n) is not convergent.
(C)	(x_n) is convergent and (y_n) is not convergent.
(D)	(x_n) is not convergent and (y_n) is convergent.
Q.2	For each positive integer n, define $x_n = (-1)^n$. Which ONE of the following statements about the sequence (x_n) is FALSE?
(A)	There exists $\epsilon > 0$ such that $\lvert x_n - 1\rvert < \epsilon$ for all positive integers n.
(B)	There exists $\epsilon > 0$ such that for all $M > 0$ there exists a positive integer $N > M$ for which $\lvert x_N - 1\rvert > \epsilon$.
(C)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $\lvert x_N - 1\rvert < \epsilon$.
(D)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $\lvert x_N - 1\rvert > \epsilon$.

Q.3	Let $f:(1,4) \to \mathbb{R}$ be a differentiable function such that $$f'(x) = (f(x) - \pi x)^2 + \pi \qquad \text{for all } x \in (1,4).$$ Which ONE of the following is a possible value of $f(3) - f(2)$?
(A)	$\pi + \frac{1}{6}$
(B)	$\pi - \frac{1}{6}$
(C)	$\frac{\pi}{2} + \frac{1}{3}$
(D)	$\frac{\pi}{2} + \frac{1}{2}$

Q.4	The general solution of the differential equation $$x \sin\left(\frac{y}{x}\right)\frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + \frac{x}{2}, \quad x > 0$$ is given by ______.
(A)	$\cos\left(\frac{x}{y}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(B)	$\cos\left(\frac{y}{x}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(C)	$\cos\left(\frac{x}{y}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant
(D)	$\cos\left(\frac{y}{x}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant

Q.5	Let $f:\mathbb{R}^3 \to \mathbb{R}$ be a continuous function. Then $$\int_0^2\int_0^x\int_0^y f(x,y,z)\,dz\,dy\,dx \;=\; \_\_\_\_\_.$$
(A)	$\int_0^2\int_0^y\int_y^2 f(x,y,z)\,dx\,dz\,dy$
(B)	$\int_0^2\int_0^x\int_z^2 f(x,y,z)\,dy\,dz\,dx$
(C)	$\int_0^2\int_y^2\int_0^x f(x,y,z)\,dz\,dx\,dy$
(D)	$\int_0^2\int_z^2\int_0^x f(x,y,z)\,dy\,dx\,dz$

Q.6	Let $f:(1,2)\to\mathbb{R}$ be a continuous function satisfying $$\lim_{x\to 1+} f(x) = 3 \text{ and } \lim_{x\to 2-} f(x) = 3.$$ Then which ONE of the following is necessarily TRUE?
(A)	f is bounded and f has a maximum or a minimum but not both.
(B)	f is bounded and f has a maximum or a minimum or both.
(C)	f is bounded and f has neither a maximum nor a minimum.
(D)	f is unbounded and f has either no maximum or no minimum.
Q.7	Let $f:\mathbb{R}\to\mathbb{R}$ be a twice differentiable function such that $f(n)=1$ for all $n\in\mathbb{Z}$. Which ONE of the following statements is necessarily TRUE?
(A)	$f'(n)=0$ for all $n\in\mathbb{Z}$.
(B)	$f'(x)=0$ for infinitely many $x\in\mathbb{R}\setminus\mathbb{Z}$.
(C)	$f''(n)=0$ for all $n\in\mathbb{Z}$.
(D)	$f''(x)=0$ for infinitely many $x\in\mathbb{R}\setminus\mathbb{Z}$.

Q.8	Let A be an invertible 5×5 real matrix. Let B be the matrix obtained by interchanging the second and third columns of A and then adding 3 times the third column to the fifth column. Which ONE of the following statements is TRUE?
(A)	B^{-1} is obtained by interchanging the second and third rows of A^{-1}, and then adding 3 times the third row to the fifth row.
(B)	B^{-1} is obtained by interchanging the second and third rows of A^{-1}, and then adding -3 times the fifth row to the third row.
(C)	B^{-1} is obtained by adding -3 times the third row to the fifth row, and then interchanging the second and third rows of A^{-1}.
(D)	B^{-1} is obtained by adding 3 times the fifth row to the third row, and then interchanging the second and third rows of A^{-1}.
Q.9	Which ONE of the following statements is FALSE?
(A)	S_3 is isomorphic to a subgroup of S_4.
(B)	$\mathbb{Z}_3$ is isomorphic to a subgroup of S_4.
(C)	S_3 is isomorphic to a quotient group of S_4.
(D)	$\mathbb{Z}_6$ is isomorphic to a quotient group of S_4.

Q.10	Let T be the triangular region in the plane with vertices at $(0,0)$, $(1,0)$ and $(1,1)$. Let f be a function defined from T to $\mathbb{R}$ given in polar coordinates. Then the double integral of f over T is equal to ______.
(A)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\ d\theta\right) r\,dr + \int_1^{\sqrt{2}}\left(\int_0^{\cos^{-1}(1/r)} f\ d\theta\right) rdr$
(B)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\ d\theta\right) rdr - \int_1^{\sqrt{2}}\left(\int_0^{\cos^{-1}(1/r)} f\ d\theta\right) rdr$
(C)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\ d\theta\right) r\,dr + \int_1^{\sqrt{2}}\left(\int_0^{\sin^{-1}(1/r)} f\ d\theta\right) rdr$
(D)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\ d\theta\right) r\,dr - \int_1^{\sqrt{2}}\left(\int_0^{\sin^{-1}(1/r)} f\ d\theta\right) rdr$

Section A: Q.11 – Q.30 Carry TWO marks each.	
Q.11	Given a sequence of real numbers (a_n), define $b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0 \end{cases}$ and $c_n = \begin{cases} a_n, & \text{if } a_n < 0 \\ 0, & \text{if } a_n \geq 0 \end{cases}$ for each positive integer n. Which ONE of the following statements is TRUE?
(A)	If $\sum_{n=1}^{\infty} a_n$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(B)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} \lvert a_n \rvert$ does not converge, then $\sum_{n=1}^{\infty} b_n$ converges.
(C)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} \lvert a_n \rvert$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(D)	If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges or $\sum_{n=1}^{\infty} c_n$ converges.

Q.12	For which one of the following pairs of values of a and b, does the series $\sum_{k=1}^{\infty} \frac{(3x)^k}{2^{\sqrt{k}}(1-x)^k}$ converge for all $x \in (a, b)$?
(A)	$a = \frac{-1}{4}$ and $b = \frac{1}{2}$
(B)	$a = -1$ and $b = \frac{1}{5}$
(C)	$a = \frac{-1}{2}$ and $b = \frac{1}{4}$
(D)	$a = \frac{-1}{4}$ and $b = \frac{1}{3}$
Q.13	Let $f: \mathbb{R} \to \mathbb{R}$ be a function that has derivatives of all orders. Which ONE of the following statements is FALSE?
(A)	If f is a polynomial in x of degree at most 5, then $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$.
(B)	If $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$, then f is a polynomial in x of degree at most 5.
(C)	If $f^{(k)}(a) = 0$ for all $1 \le k \le 5$ and $f^{(6)}(a) > 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local minimum at a.
(D)	If $f^{(k)}(a) = 0$ for all $1 \le k \le 6$ and $f^{(7)}(a) < 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local maximum at a.

Q.14	Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\begin{pmatrix}4\\5\\6\end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?
(A)	$\begin{pmatrix}3\\3\\3\end{pmatrix}$
(B)	$\begin{pmatrix}5\\7\\9\end{pmatrix}$
(C)	$\begin{pmatrix}5\\6\\9\end{pmatrix}$
(D)	$\begin{pmatrix}7\\8\\9\end{pmatrix}$

Q.15	Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function given by $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2$ for all $(x, y) \in \mathbb{R}^2$. Which ONE of the following statements is TRUE?
(A)	f has local maxima at exactly two points.
(B)	f has local minima at exactly two points.
(C)	f has local minima at exactly three points.
(D)	f has exactly four saddle points.
Q.16	The orthogonal trajectories of the family of curves $y = -3x - 3 + me^x$ for a real parameter m are given by _______.
(A)	$x = \frac{y}{3} - \frac{1}{9} + k\, e^{-3y}$, k is a real parameter
(B)	$x = \frac{y}{3} - \frac{1}{9} + k\, e^{3y}$, k is a real parameter
(C)	$x = \frac{-y}{3} + \frac{1}{9} + k\, e^{-3y}$, k is a real parameter
(D)	$x = \frac{-y}{3} + \frac{1}{9} + k\, e^{3y}$, k is a real parameter

Q.17	Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by $$f(x,y) = \begin{cases} \dfrac{5x^2y - 4y^3}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$ Which ONE of the following statements is FALSE?
(A)	f is continuous at $(0,0)$
(B)	$\dfrac{\partial f}{\partial x}$ at $(0,1) = 0$
(C)	$\dfrac{\partial f}{\partial y}$ at $(1,0) = 5$
(D)	$\dfrac{\partial f}{\partial y}$ is continuous at $(0,0)$

Q.18	Let $\phi: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ be the solution of the differential equation $$(\cos x)\frac{dy}{dx} - y = 2y^2(\sin x - 1)\cos x$$ satisfying $\phi(0) = \frac{1}{2}\cdot$ Then $\phi\left(\frac{\pi}{4}\right) =$______.
(A)	2
(B)	$\dfrac{3+2\sqrt{2}}{2}$
(C)	$\dfrac{1}{2}$
(D)	$\dfrac{1}{\sqrt{2}}$

Q.19	The value of the following expression $$\lim_{x\to\infty}\int_0^x e^{-t}\lvert\sin t\rvert\,dt$$ is _____.
(A)	$\dfrac{e^{\pi}+1}{2(e^{\pi}-1)}$
(B)	$\dfrac{e^{\pi}-1}{2(e^{\pi}+1)}$
(C)	$\dfrac{2(e^{\pi}+1)}{e^{\pi}-1}$
(D)	$\dfrac{2(e^{\pi}-1)}{e^{\pi}+1}$
Q.20	The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by the angle $\frac{\pi}{2}$ in the plane about the origin. What is the area of the planar region traced out by the semi-circle?
(A)	6π
(B)	4π
(C)	2π
(D)	π

Q.21	Let $f: [1, 9] \to \mathbb{R}$ be a non-constant continuous function such that $f(1) = f(9)$. Which ONE of the following statements is necessarily TRUE?
(A)	There exists $c \in [4, 8]$ such that $f(c) = f(c + 1)$.
(B)	There exists $c \in [3, 7]$ such that $f(c) = f(c + 2)$.
(C)	There exists $c \in [2, 6]$ such that $f(c) = f(c + 3)$.
(D)	There exists $c \in [1, 5]$ such that $f(c) = f(c + 4)$.
Q.22	Let S be the solid of intersection of two solid spheres S_1, S_2 given by $S_1: x^2 + y^2 + (z - 1)^2 \leq 4$ and $S_2: x^2 + y^2 + (z + 1)^2 \leq 4$. Which ONE of the following iterated integrals expresses the volume of S?
(A)	$8\int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} \left(\sqrt{4 - x^2 - y^2} - 1\right) dy\, dx$
(B)	$8\int_0^{2} \int_0^{\sqrt{4-x^2}} \left(\sqrt{4 - x^2 - y^2} - 1\right) dy\, dx$
(C)	$2\int_0^{2} \int_0^{\sqrt{4-x^2}} \left(\sqrt{4 - x^2 - y^2} - 1\right) dy\, dx$
(D)	$2\int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} \left(\sqrt{4 - x^2 - y^2} - 1\right) dy\, dx$

Q.23	Consider the differential equation $$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin(e^x).$$ Which ONE of the following is a particular solution of this differential equation?
(A)	$-e^{-x}\sin(e^x)$
(B)	$-e^{-2x}\sin(e^x)$
(C)	$-e^{-x}(\sin(e^x) + \cos(e^x))$
(D)	$-e^{-2x}(\sin(e^x) + \cos(e^x))$
Q.24	Let e_1, e_2, e_3, e_4, e_5 denote the standard basis vectors of the real vector space $\mathbb{R}^5$. Let V be the subspace of $\mathbb{R}^5$ spanned by the vectors $e_1 + e_2, e_2 + e_3, e_3 + e_4 + e_5$. Consider the real vector space $M_{2,5}(\mathbb{R})$. Let $$S = \{P \in M_{2,5}(\mathbb{R}) \mid \text{nullspace}(P) = V\}.$$ Which ONE of the following statements is TRUE?
(A)	S is the empty set.
(B)	$S = \{0\}$.
(C)	S is non-empty and S is not a subspace of $M_{2,5}(\mathbb{R})$.
(D)	S is a nonzero subspace of $M_{2,5}(\mathbb{R})$.

Q.25	Let P be a nonzero complex $n \times n$ matrix such that $v^* P v \geq 0$ for all column vectors v in $\mathbb{C}^n$. Which ONE of the following statements is necessarily TRUE?
(A)	All eigenvalues of P are negative real numbers.
(B)	If v_1, v_2 are column vectors in $\mathbb{C}^n$ such that $P v_1 = v_2$ and $P v_2 = v_1$, then $v_1 = v_2$.
(C)	$v^* P v \neq 0$ for all nonzero column vectors v in $\mathbb{C}^n$.
(D)	All eigenvalues of P^* are negative real numbers.
Q.26	Let $f: G \longrightarrow \mathbb{Z}$ be a surjective group homomorphism from an abelian group G to the additive group of integers $\mathbb{Z}$. Let K denote the kernel of f. Which ONE of the following statements is FALSE?
(A)	There exists a group homomorphism $\phi: \mathrm{Z} \longrightarrow G$ such that $\mathrm{f} \circ \phi$ is the identity homomorphism.
(B)	K is isomorphic to a quotient group of G.
(C)	For all non-zero homomorphisms $\alpha: \mathrm{Z} \longrightarrow G$ and $\beta: \mathrm{G} \longrightarrow K$, the composition $\beta \circ \alpha$ is non-zero.
(D)	G has a subgroup isomorphic to $\mathbb{Z}$.

Q.27	Which ONE of the following statements is FALSE?
(A)	There is a surjective group homomorphism from the additive group of rational numbers to the multiplicative group of all complex roots of unity.
(B)	The multiplicative group of complex numbers of modulus one is isomorphic to a quotient group of the additive group of real numbers.
(C)	Any group homomorphism from the multiplicative group of nonzero complex numbers into the group of all invertible 2×2 matrices with real entries has nontrivial kernel.
(D)	There exists a group homomorphism from the symmetric group on n symbols into the multiplicative group of all invertible $n \times n$ matrices with real entries, which has a trivial kernel.

Q.28	For any two points $P, Q \in \mathbb{R}^3$, let $vect(P, Q)$ denote the vector from the point P to the point Q and let $d(P, Q)$ denote the length of $vect(P, Q)$. Let $F: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation such that $$d\big(F(P), F(Q)\big) = d(P, Q) \quad \text{for all } P, Q \in \mathbb{R}^3.$$ Which ONE of the following statements is FALSE ?
(A)	F is an injective map.
(B)	F is a surjective map.
(C)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $vect\big(F(P), F(Q)\big) \cdot vect\big(F(R), F(S)\big) = vect(P, Q) \cdot vect(R, S)$, where $\cdot$ denotes the usual dot product in $\mathbb{R}^3$.
(D)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $vect\big(F(P), F(Q)\big) \times vect\big(F(R), F(S)\big) = vect(P, Q) \times vect(R, S)$, where $\times$ denotes the usual cross product in $\mathbb{R}^3$.

Q.29	For each positive integer n, let $$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log_e n$$ and $$y_n = \int_1^n \frac{\cos t}{t^2}\, dt\ .$$ Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is convergent and (y_n) is not convergent.
(C)	(x_n) is not convergent and (y_n) is convergent.
(D)	(x_n) is not convergent and (y_n) is not convergent.
Q.30	Let G be the power set of $\{1,2,3,4\}$. Let the binary operation Δ on G be $$A \,\Delta\, B = (A \setminus B) \cup (B \setminus A).$$ Which ONE of the following statements is TRUE?
(A)	$\{1\}$ is an identity element of (G, Δ).
(B)	(G, Δ) is an abelian group but not cyclic.
(C)	(G, Δ) is a group and has an element of order 4.
(D)	(G, Δ) is a group and has an element of order 8.

Section B: Q.31 – Q.40 Carry TWO marks each.	
Q.31	Which of the following functions $f: \mathbb{R} \to \mathbb{R}$ has/have a local minimum at $x = 0$?
(A)	$f(x) = \sin \lvert x \rvert$
(B)	$f(x) = \sin x + \frac{x^3}{6}$
(C)	$f(x) = x^4 + x^2 + 3$
(D)	$f(x) = \min \{ x - [x], 1 - x + [x] \}$
Q.32	Which of the following statements is/are TRUE?
(A)	There exists a monotone sequence that does not converge but has a convergent subsequence.
(B)	There exists a sequence that has a bounded subsequence but does not have any convergent subsequence.
(C)	There exists a sequence (x_n) such that given any positive integer m, (x_n) has a subsequence converging to m.
(D)	There exists a sequence (x_n) such that $(\lvert x_{n+1} - x_n \rvert)$ converges to 0 but (x_n) does not converge.

Q.33	Which of the following functions is/are differentiable at $x = 1$?
(A)	$f(x) = \|x-1\|^3$
(B)	$f(x) = \|x^2-1\|$
(C)	$f(x) = \begin{cases} x^2 e^{-x^2}, & \|x\| \le 1 \\ e^{-1}, & \|x\| > 1 \end{cases}$
(D)	$f(x) = [x]$
Q.34	Let $f: \mathbb{R} \to \mathbb{R}$ be the function given by $$f(x) = \begin{cases} \cos x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$ Which of the following statements is/are TRUE?
(A)	$f(x)$ is continuous at 0.
(B)	$f(x)$ is continuous at $\frac{\pi}{2}$.
(C)	$f(x)$ is Riemann integrable on [0, 1] and $\int_0^1 f(x)\,dx = \sin 1$.
(D)	$f(x)$ is Riemann integrable on [0, 1] and $\int_0^1 f(x)\,dx = 0$.

Q.35	Consider the differential equation $$(2y\cos x - xy\ \sin x)\,dx + 2x\cos x\,dy = 0 \quad \text{for } x \in \left(0, \frac{\pi}{4}\right).$$ Which of the following is/are integrating factor(s) of the differential equation?
(A)	$\frac{1}{xy}$
(B)	xy
(C)	$\sec x$
(D)	$\sqrt{\sec x}$
Q.36	Let $f(x,y) = \begin{cases} \frac{x^2 y}{1+x^2}\sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ Which of the following statements is/are TRUE?
(A)	$\lim_{y\to 0}\lim_{x\to 0} f(x,y)$ exists.
(B)	$\frac{\partial f}{\partial x}$ is continuous at the point $(0,1)$.
(C)	$\frac{\partial f}{\partial y}$ is continuous at the point $(0,1)$.
(D)	$\lim_{(x,y)\to(0,0)} \frac{f(x,y)}{\sqrt{x^2+y^2}}$ does not exist.

Q.37	Let $P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in M_5(\mathbb{C})$. Which of the following statements is/are TRUE?
(A)	$\text{nullity}(P - I) \geq 2$, where I is the 5×5 identity matrix.
(B)	P has 4 distinct eigenvalues in $\mathbb{C}$.
(C)	P has 3 distinct eigenvalues in $\mathbb{R}$.
(D)	If λ is an eigenvalue of P, then there exists a positive integer n such that $\lambda^n = 1$.

Q.38	Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Which of the following statements is/are TRUE?
(A)	$\sum_{k=0}^{6} (-1)^k \binom{13}{2k} = -64.$
(B)	$\sum_{k=0}^{6} (-1)^k \binom{13}{2k+1} = 128.$
(C)	$\sum_{k=0}^{6} (-1)^k \binom{13}{2k} = 64.$
(D)	$\sum_{k=0}^{6} (-1)^k \binom{13}{2k+1} = -128.$
Q.39	Consider matrices P of order 4×6 and Q of order 6×4 with real entries such that $PQ = 0$ and $QP = 0$. Which of the following statements is/are TRUE?
(A)	$\text{rangespace}(P) \subseteq \text{nullspace}(Q)$ and $\text{rangespace}(Q) \subseteq \text{nullspace}(P)$.
(B)	$\text{rank}(P) + \text{rank}(Q) \leq 4.$
(C)	If $\text{rangespace}(P) = \text{nullspace}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4.$
(D)	$\text{rangespace}(Q) = \text{nullspace}(P).$

Q.40	Let V be the subset of $\mathbb{R}$ defined by $$V = \left\{\frac{a + b\sqrt{2}}{c + d\sqrt{2}} : a, b, c, d \in \mathbb{Q}, c^2 + d^2 \neq 0\right\}.$$ Which of the following statements is/are FALSE?
(A)	V is closed under the usual addition in $\mathbb{R}$.
(B)	V is a subspace of the real vector space $\mathbb{R}$.
(C)	V is a two dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
(D)	V is a four dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
Section C: Q.41 – Q.50 Carry ONE mark each.	
Q.41	The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} (\log_e n)^{-1} x^n$ is _______ (*rounded off to one decimal place*).
Q.42	$\int_0^1 \left(\sum_{k=1}^{\infty} \frac{(\log_e 2)^k x^{k-1}}{(k-1)!}\right) dx =$ ______(*rounded off to one decimal place*).

Q.43	$\lim_{n\to\infty}\left(\frac{n^2+1}{\sqrt{n^6+1}}+\cdots+\frac{n^2+n}{\sqrt{n^6+n}}\right)=$_____(*rounded off to one decimal place*).
Q.44	$\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{\sin x}+e^{\frac{1-\cos x}{x}}\right)=$ ______(*rounded off to one decimal place*).
Q.45	Let α and β be real numbers such that the differential equation $(y^3+\alpha x y^4-5x+\cos 2y)dx+(3xy^2+20x^2y^3+\beta x\sin 2y)dy=0$ is exact. Then $\alpha+\beta=$ ______(*rounded off to one decimal place*).
Q.46	Let $z=\cos(4x+5y)$, where $x=\frac{\pi}{2}+2\theta,\ y=-\left(\frac{\pi}{4}+\theta\right)$. Then $\left.\frac{dz}{d\theta}\right\vert_{\theta=\frac{\pi}{4}}=$ ______(*rounded off to one decimal place*).

Q.47	Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30}, then the total number of such distinct surjective homomorphisms is ______.
Q.48	$\int_0^3 (\,\lvert x-1\rvert - x[x]\,)\,dx =$ ______(*rounded off to one decimal place*).
Q.49	Let $A \in M_3(\mathbb{C})$. Suppose the column vector $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix}$ in $\mathbb{C}^3$ belongs to the intersection of nullspace(A) and rangespace(A^T). Then $\lvert x\rvert =$ ______(*rounded off to one decimal place*).
Q.50	Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P. Then $\det(Q) =$ ________ (*rounded off to one decimal place*).

Section C: Q.51 – Q.60 Carry TWO marks each.	
Q.51	Let α, β and γ be fixed real numbers such that $y(x) = C_1 e^{-x} + C_2 e^{2x} + \alpha x e^{-x}$, where C_1 and C_2 are arbitrary real constants, is the general solution of the differential equation $$\frac{d^2y}{dx^2} + \beta \frac{dy}{dx} + \gamma y = -e^{-x}.$$ Then $\alpha(\beta + \gamma)$= ______(*rounded off to one decimal place*).
Q.52	Let the function $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $$f(x,y) = \begin{cases} 4x^2 \tan^{-1}\left(\frac{y}{2x}\right) + y^3 \tan^{-1}\left(\frac{x}{4y^2}\right), & xy \neq 0 \\ 0, & \text{otherwise} \end{cases}$$ Then $\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\right)$ at $(0,0)$ is ______ (*rounded off to two decimal places*).
Q.53	The double integral of $f(x,y) = x$ over the triangular region with vertices at $\left(-\frac{1}{2}, \frac{1}{2}\right)$, $(1,2)$ and $(1,-1)$ is ______(*rounded off to one decimal place*).

Q.54	Let $$f(x) = \int_0^{e^{\sin x}} \int_0^{\log_e y} e^{-\left(\frac{1}{\sqrt{2}}\right)} (1 - t^2)\, dt\, dy.$$ Then $f'(\pi/4) =$ ______*(rounded off to one decimal place)*.
Q.55	The volume of the tetrahedron bounded by the planes $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is ______*(rounded off to one decimal place)*.
Q.56	Let $M \in M_3(\mathbb{R})$. If $$\mathrm{M}\begin{pmatrix}\sqrt{15}\\ \sqrt{15}\\ 0\end{pmatrix} = \begin{pmatrix}\sqrt{3}\\ \sqrt{2}\\ -5\end{pmatrix}, \quad M\begin{pmatrix}0\\ 0\\ \sqrt{30}\end{pmatrix} = \begin{pmatrix}\sqrt{15}\\ \sqrt{10}\\ \sqrt{5}\end{pmatrix}, \quad M\begin{pmatrix}-\sqrt{15}\\ \sqrt{15}\\ 0\end{pmatrix} = \begin{pmatrix}\sqrt{12}\\ -\sqrt{18}\\ 0\end{pmatrix}$$ then $\det(M) =$ ______*(rounded off to one decimal place)*.

Q.57	Let $a, b \in \mathbb{R}$. If the system of linear equations $$x + 2y + 2z = 1$$ $$2x + 3y + z = 2$$ $$ax + 5y + bz = b$$ has infinitely many solutions, then $a + b =$ ______ (*rounded off to one decimal place*).
Q.58	A fruit shop has 4 different types of bananas. The number of ways in which 12 bananas can be bought with at least one banana from each type, is ______.
Q.59	Let A be a 3×3 real matrix such that given any column vector $x \in \mathbb{R}^3$, the column vector Ax is the reflection of x about the plane $$\{(a, b, -a-b) : a, b \in \mathbb{R}\}.$$ Then the sum of the diagonal elements of A is ______(*rounded off to one decimal place*).

Q.60	Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Then, $$\binom{5}{0}+\binom{6}{1}+\binom{7}{2}+\binom{8}{3}+\binom{9}{4}+\binom{10}{5}+\binom{11}{6}= \_\_\_\_\_.$$

JAM 2026: Mathematics (MA)
Master Answer Key

Question No.	Subject	Question Type	Answer Key/Range	Mark(s)
1	MA	MCQ	D	1
2	MA	MCQ	D	1
3	MA	MCQ	A	1
4	MA	MCQ	D	1
5	MA	MCQ	A	1
6	MA	MCQ	B	1
7	MA	MCQ	B	1
8	MA	MCQ	B	1
9	MA	MCQ	D	1
10	MA	MCQ	MTA	1
11	MA	MCQ	C	2
12	MA	MCQ	C	2
13	MA	MCQ	D	2
14	MA	MCQ	D	2
15	MA	MCQ	D	2
16	MA	MCQ	C	2
17	MA	MCQ	D	2
18	MA	MCQ	D	2
19	MA	MCQ	A	2
20	MA	MCQ	A	2
21	MA	MCQ	D	2
22	MA	MCQ	A	2
23	MA	MCQ	B	2
24	MA	MCQ	C	2
25	MA	MCQ	B	2
26	MA	MCQ	C	2
27	MA	MCQ	C	2
28	MA	MCQ	D	2
29	MA	MCQ	A	2
30	MA	MCQ	B	2

Question No.	Subject	Question Type	Answer Key/Range	Mark(s)
31	MA	MSQ	A;C;D	2
32	MA	MSQ	C;D	2
33	MA	MSQ	A;C	2
34	MA	MSQ	B	2
35	MA	MSQ	B;D	2
36	MA	MSQ	A;C	2
37	MA	MSQ	A;B;D	2
38	MA	MSQ	A	2
39	MA	MSQ	A;B;C	2
40	MA	MSQ	B;D	2
41	MA	NAT	4.0 to 4.0	1
42	MA	NAT	1.0 to 1.0	1
43	MA	NAT	1.0 to 1.0	1
44	MA	NAT	1.0 to 1.0	1
45	MA	NAT	8.0 to 8.0	1
46	MA	NAT	3.0 to 3.0	1
47	MA	NAT	8 to 8	1
48	MA	NAT	-4.0 to -4.0	1
49	MA	NAT	3.0 to 3.0	1
50	MA	NAT	16.0 to 16.0	1
51	MA	NAT	-1.0 to -1.0	2
52	MA	NAT	0.24 to 0.26	2
53	MA	NAT	1.1 to 1.3	2
54	MA	NAT	0.4 to 0.5	2
55	MA	NAT	4.0 to 4.0	2
56	MA	NAT	-1.0 to -1.0	2
57	MA	NAT	6.0 to 6.0	2
58	MA	NAT	165 to 165	2
59	MA	NAT	1.0 to 1.0	2
60	MA	NAT	924 to 924	2